

Do Mergers and Acquisitions Improve Efficiency? Evidence from Power Plants

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We analyze acquisitions in US power plants by using rich data on hourly productivity and thousands of ownership changes. We find a 2% average increase in efficiency for acquired plants, beginning 5 months after acquisition. Efficiency gains rise to 5% under direct ownership changes, with no significant change when only parent ownership changes. Investigating the mechanisms, three-quarters of the efficiency gain is attributed to increased productive efficiency, while the rest comes from dynamic efficiency through changes in production allocation. Our evidence suggests that high-productivity firms buy underperforming assets from low-productivity firms and make them as productive as their existing assets through operational improvements.

I. Introduction

A fundamental issue in antitrust policy is the trade-off between the market power and efficiency effects of mergers. The increase in market power raises prices for consumers; however, potential efficiency gains can counteract this

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effect, rendering the net impact of mergers on welfare ambiguous (Williamson 1968). Although there is extensive literature on the price effects of mergers, we have limited evidence on how mergers affect efficiency (Whinston 2007; Asker and Nocke 2021).¹ With little guidance from empirical evidence, researchers analyzing the competitive effects of mergers often rely on hypothetical efficiency gains (Farrell and Shapiro 2010; Nocke and Whinston 2022; Berger et al. 2023).²

A major challenge in analyzing the efficiency effects of mergers is distinguishing true efficiency gains from other potential merger effects, such as changes in market power, buyer power, and product quality. Because of the limitations of common production datasets, most productivity studies rely on revenue-based total factor productivity (TFPR), derived from revenues and input expenditures, rather than quantity-based measures (Foster, Haltiwanger, and Syverson 2008; Atalay 2014). Using TFPR is particularly problematic in merger analysis because changes in market power, buyer power, or quality can affect TFPR even in the absence of efficiency gains. This makes it difficult to identify the true efficiency effects of mergers.³

In this paper, we provide large-scale evidence on the efficiency effects of mergers while tackling these challenges. We focus on acquisitions in the US electricity generation industry between 2000 and 2023. Four key features of this industry allow us to overcome the difficulties in quantifying merger efficiencies. First, we observe, at the hourly frequency, the physical quantities of both output and the primary input, the consumption of fuel, which makes up 79% of operational costs. Using these high-frequency data, we construct an efficiency measure (heat rate) and analyze how it changes around the time of acquisition. Second, electricity is a homogeneous product, eliminating potential quality changes that could confound our analysis. Third, the power generation industry experienced a significant number of acquisitions during the sample period. Our sample includes 505 transactions with 3,515 generator ownership changes, representing an average of 4.5% of the industry's annual capacity. These ownership changes exhibit significant heterogeneity in transaction, firm, and

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¹ Weinberg (2008), Ashenfelter and Hosken (2010), and Kwoka (2014) provide reviews of the literature on the price effects of mergers.

² Consider, e.g., these quotes from Nocke and Whinston (2022): “there is a clear need for much better evidence on the efficiency effects” (1946); “we observe that the literature on efficiency effects of horizontal mergers is extremely limited” (1918); and “there is remarkably little solid empirical evidence on this point” (1945).

³ The examination of efficiencies is a standard part of merger review (see DOJ and FTC 2023, sec. 3.3).

plant characteristics, which we leverage to study the mechanisms of efficiency gains. Finally, electricity generation is an important industry, with efficiencies leading to positive externalities through decreased emissions (EPA 2018).

Our analysis begins with a difference-in-differences estimation, comparing the efficiency of acquired plants to those not involved in acquisitions. We find that, on average, the efficiency of acquired plants increases by 2.0% after acquisitions. However, this average effect masks significant heterogeneity from the two types of ownership changes observed in the data: parent- and subsidiary-level ownership changes. Ownership changes only at the parent level do not change efficiency on average, whereas changes in subsidiary ownership result in a 4.9% increase. This finding highlights the influence of the direct owner in power plant operations. Finally, examining the timing of efficiency gains, we observe improvements beginning 5 months after acquisition, stabilizing after 18 months. This suggests that new owners require time to implement efficiency-improving changes.

Efficiency increases in electricity generation can manifest in various ways, not all of which are necessarily welfare improving. For instance, generators incur additional costs when adjusting production levels, known as ramp costs, so acquirers might improve efficiency by reducing production and ramping (Borrero, Gowrisankaran, and Langer 2024). Alternatively, they can use generators more intensively at the expense of reliability (Borenstein, Bushnell, and Mansur 2023). To understand the nature of efficiency effects, we study other generator outcomes indicative of performance, including output, capacity utilization, outages, and emissions. We find that acquired generators increase generation by 7.3%, raise capacity utilization by 2.2%, and experience a 33.5% reduction in outages following the acquisition. These results suggest that acquirers improve other dimensions of plant performance beyond efficiency and efficiency gains occur without deterioration in other performance indicators.

While evidence of efficiency gains after mergers is important, understanding the underlying mechanisms is essential for informing antitrust policy and generalizing findings from this industry to others. With this motivation, the remainder of the paper conducts a comprehensive mechanism analysis by investigating heterogeneity in the efficiency effects and modeling sources of efficiency gains in power plants.

We start by analyzing the characteristics of generators, firms, and transactions that may be informative about efficiency effects. We find that, on average, generators with above-median capacity experience a 3.3 percentage point larger efficiency increase than those with below-median capacity. This difference perhaps reflects greater incentives to improve efficiency in larger-capacity units, as any improvement in efficiency would yield higher returns. Regarding firm characteristics, efficiency improvement is 4.1 percentage points higher when the acquirer is larger than the median and

5.8 percentage points higher when the acquirer is a serial acquirer. These results suggest that a firm's experience in plant operation and acquisition is an important predictor of postacquisition efficiency gains. Finally, we examine the differential impact of cross-market acquisitions on efficiency, finding that they result in efficiency gains 3.9 percentage points lower than within-market acquisitions.

We then proceed to a more structural analysis to uncover the mechanisms behind efficiency gains. We focus on two mechanisms through which a firm can increase the efficiency of electricity generation: (i) improving the performance of individual generators (productive efficiency) and (ii) optimizing production allocation dynamically within a generator to reduce ramping (dynamic efficiency). We develop a testable prediction for each mechanism and quantify their contributions by modeling the efficiency of generators. In particular, relying on a Leontief electricity production function as a microfoundation, we model the heat rate as a function of output, ramp rate, and weather conditions (temperature and humidity). The availability of hourly production data allows us to estimate this function for each generator separately for pre- and postacquisition periods, thereby directly measuring the change in production function due to acquisitions.

We test the role of the first mechanism, productive efficiency, by quantifying the efficiency change due to shifts in the heat rate curve, controlling for ramp and weather conditions. For dynamic efficiency, we analyze the variation of an acquired plant's production, with less variation over time indicating dynamic efficiency. To quantify its contribution, we use the estimated production function model and calculate the efficiency increase due to changes in the postacquisition production distribution while keeping the heat rate curve as in the preacquisition technology.

We find that productive efficiency accounts for most (three-quarters) of the total efficiency gain. The average heat rate curve of acquired generators shifts downward after acquisition at every production level, suggesting that acquirers improve generators' internal efficiency. We also find evidence supporting an increase in dynamic efficiency. Following acquisitions, generators' coefficient of variation (CV) of production tends to decrease, explaining the remainder of the efficiency increase.

Having established the role of productive efficiency, the next natural question is how acquirers improve productive efficiency. There are two potential channels: (i) operational improvements, which involve, for example, installing control software, implementing effective maintenance, providing personnel training, or adopting best practices, and (ii) capital investments, which involve equipment upgrades. Process improvements suggest transfers of intangible capital after acquisition (Atalay, Hortaçsu, and Syverson 2014), whereas capital upgrades indicate either credit constraints faced by the former owner or insufficient incentives to make

efficiency-improving capital investments (Midrigan and Xu 2014). Although the large-scale nature of our study precludes us from directly observing what changes inside the plant, we supplement our efficiency data with two additional datasets to distinguish between these mechanisms: (i) data on plant managers and (ii) data on capital expenditures and non-fuel costs.

Starting with the manager data, we find that 55% of acquired power plants change managers within 1 year of acquisition. These new managers are 5.6 percentage points more likely to have a master's degree and 4.3 percentage points more likely to have a bachelor's degree than managers of nonacquired plants. In contrast, we find no evidence of increased capital expenditures or nonfuel costs after acquisitions. Therefore, the new owners appear to improve efficiency through low-cost operational improvements rather than high-cost capital investments. This analysis contributes to the growing body of evidence suggesting that acquisitions serve as a mechanism for transferring within-organization knowledge to newly acquired assets (Hortaçsu and Syverson 2007; Bloom, Sadun, and Van Reenen 2012; Atalay, Hortaçsu, and Syverson 2014; Eliason et al. 2020).

Efficiency gains through operational improvements point to superior capabilities of acquirers in plant operation and utilization compared to target firms. To further explore this and understand how acquisitions reallocate assets within the economy, our final analysis estimates and compares the productivity of target and acquirer firms. We find that high-productivity firms buy underperforming assets from low-productivity firms and make the acquired assets as productive as their existing assets after acquisition. On average, acquirers are 1.7% more productive than the targets, and assets sold by the target firms underperform their other assets by 3%. These findings suggest that acquisitions allocate assets to firms with both relative and absolute advantages in utilizing those assets, providing evidence for both the "high-buys-low" and "like-buys-like" theories of merger gains in the literature (Jovanovic and Rousseau 2002; Rhodes-Kropf and Robinson 2008).

As with all retrospective merger analyses, an identification challenge in our paper is the potential endogeneity of mergers. To address this concern, we implement several strategies and robustness checks. First, our specification incorporates a rich set of controls along with flexible time trends (fuel, technology, vintage, and state), accounting for factors that potentially influence selection into acquisitions. Second, we analyze the timing of the effect, demonstrating parallel trends between the treated and control groups 3 years before acquisitions and an increase in efficiency starting a few months after acquisitions. Third, we show that our results are robust to the empirical method, sample period, acquisition definition, and data frequency in measuring efficiency.

We conclude the introduction by noting that our results do not characterize the full impact of acquisitions, as we have identified only one component of the merger welfare analysis. Furthermore, the magnitude of the efficiency effect identified in this paper might not extend to industries with production techniques or acquisition motives that differ significantly from those in electricity generation. Although we focus on a single sector to leverage the available data and numerous acquisitions, we provide detailed evidence of mechanisms to draw broader lessons from this sector.

Contribution to the literature.—This article contributes to the literature on the effects of mergers on productivity. As noted by Whinston (2007) and Asker and Nocke (2021) in the two latest industrial organization handbooks, there is a limited number of papers on the productivity effects of mergers. Among these studies, Blonigen and Pierce (2016) apply the De Loecker and Warzynski (2012) method to separately identify the effects of mergers on market power and productivity in US manufacturing plants. Their findings suggest an increase in market power but no evidence of a productivity effect. Kulick (2017) studies mergers in the US ready-mix concrete industry, finding that prices rose because of increased market power despite a 6% productivity increase in acquired plants. Braguinsky et al. (2015) examine the effects of consolidation in the early twentieth-century Japanese cotton spinning industry. They find that although acquirers are not more productive conditional on operation, they are more profitable because of better inventory management and higher capacity utilization. After acquisitions, the acquirers improve capacity utilization in the acquired plants, raising both productivity and profitability.⁴

This article contributes to the literature studying efficiency in the power generation industry, which has focused mainly on the effects of deregulation that began in the 1990s (Knittel 2002; Bushnell and Wolfram 2005; Fabrizio, Rose, and Wolfram 2007; Davis and Wolfram 2012; Hausman 2014; Cicala 2015, 2022). These studies compare the performance of plants in states that pursued restructuring to those in states that did not, generally finding a positive impact of restructuring on plant

⁴ Evidence of cost savings from other industries includes meat products (Nguyen and Ollinger 2006), railroads (Bitzan and Wilson 2007; Chen 2024), electricity distribution (Kwoka and Pollitt 2010; Clark and Samano 2022), radio (Jeziorski 2014), banking (Focarelli and Panetta 2003), and health care (Dranove and Lindrooth 2003; Harrison 2011; Schmitt 2017). These studies typically analyze firm costs, which include both input prices and firm productivity. Another strand of literature provides evidence on efficiency effects by analyzing a single merger. Some examples are the Molson-Coors merger (Grieco, Pinkse, and Slade 2018) and the Miller-Coors merger (Ashenfelter, Hosken, and Weinberg 2015) in the brewing industry and the Boeing–McDonnell Douglas merger (An and Zhao 2019) in the aerospace industry.

operations.⁵ Our paper differs from this literature, as we analyze the effects of ownership changes rather than deregulation on productivity.⁶ We primarily focus on the period after deregulation and exclude forced divestitures due to deregulation from our sample. In the literature studying electricity markets, our paper also relates to the work of Hortaçsu and Puller (2008) and Hortaçsu et al. (2019), who study the heterogeneity of firms' strategic bidding ability in wholesale electricity auctions and how ownership changes potentially affect strategic ability.

Finally, this paper contributes to the recent wave of retrospective merger research that examines the impact of mergers on various firm outcomes. This literature has advanced our understanding of cross-market mergers (Lewis and Pflum 2017; Dafny, Ho, and Lee 2019), vertical mergers (Luco and Marshall 2020), monopsony power (Prager and Schmitt 2021), buyer power (Craig, Grennan, and Swanson 2021), price effects (Brand, Garmon, and Rosenbaum 2023; Bhattacharya, Illanes, and Stillerman 2024; Brot-Goldberg et al. 2024), quality (Eliason et al. 2020; La Forgia 2024), product availability (Atalay et al. 2024), firm entry (Fan and Yang 2025), capacity utilization (Kalnins, Froeb, and Tschantz 2017), employment (Geurts and Van Biesebroeck 2019), and political influence (Moshary and Slattery 2024). We complement this body of work by studying how mergers affect efficiency and providing evidence of the mechanisms.

II. Institutional Background and Plant Productivity

This section begins with the institutional background of the power generation sector, followed by an overview of mergers and acquisitions in the industry. We then discuss power plant operation and our approach to measuring plant productivity.

A. *The Power Generation Sector in the United States*

The US electric power sector accounts for roughly 2% of US GDP (Bradley and Associates 2017). Before the 1990s, US electricity generation was predominantly supplied by regulated and vertically integrated utilities. These entities typically served a specific territory and controlled all components of the sector, including generation, transmission, and distribution. The returns of these utilities were subject to rate of return on capital investment and cost-of-service regulation (Joskow, Bohi, and Gollop

⁵ MacKay and Mercadal (2024) find that, despite decreasing generation costs, wholesale prices increased because of market imperfections.

⁶ Bushnell and Wolfram (2005) also study the impact of ownership changes on power plant efficiency. Their study focused on utility divestitures in the context of industry deregulation. By contrast, our study examines ownership changes that occurred after deregulation.

1989). This highly regulated market structure provided minimal incentives for efficiency improvements, leading to significant inefficiencies in electricity generation (Fabrizio, Rose, and Wolfram 2007; Cicala 2015).

In the 1990s, the industry underwent significant deregulation. In many states, electricity generation was decoupled from transmission and distribution, with most generators transitioning to market-based compensation. This shift coincided with the establishment of independent system operators (ISOs), which manage the electricity grid and organize the wholesale market for electricity. By 2020, about 70% of US electricity demand was served through seven ISOs (EIA 2020).⁷

B. Acquisitions in the Power Sector

Large power companies are often structured into multiple subsidiaries under a single parent company, each serving distinct locations and segments of the power sector. These parent companies frequently own assets in generation, transmission, and distribution within the same region, although some operate subsidiaries across various parts of the country. Following the deregulation wave in the 1990s, there was a notable increase in mergers and acquisitions among these entities (Davis and Wolfram 2012). Moreover, financial firms, particularly private equity firms and bank funds, began investing heavily in the power generation sector (Andonov and Rauh 2023).

Acquisitions in the power sector can be categorized into two types: (i) asset acquisitions and (ii) subsidiary acquisitions. Asset acquisitions involve a firm selling parts of its power plant portfolio, with the acquired assets placed under a subsidiary of the acquiring company. Subsidiary acquisitions occur when a parent company acquires another company's subsidiary, including all its assets. In asset acquisitions, both parent and subsidiary owners change; in subsidiary acquisitions, only the parent owner changes.⁸ For a visual explanation of these acquisition types, see figure OA-3 (figs. OA-1–OA-23 are in the online appendix).

Proposed power plant acquisitions in the US electricity sector are subject to review by the Federal Energy Regulatory Commission (FERC), the Department of Justice (DOJ), and state public utility commissions (PUCs; Niefer 2012). FERC conducts its review under Section 203 of the Federal Power Act to determine whether the merger aligns with the public interest (FERC 2012). The DOJ's review focuses on potential anticompetitive effects. If the DOJ, FERC, or a PUC finds consumer harm, they may

⁷ We use ISO as an umbrella term for both ISOs and regional transmission organizations.

⁸ In some cases, two companies merge to form a new entity, and power plants become part of this new entity. These cases typically fall under subsidiary acquisitions.

challenge the merger or require remedies.⁹ Despite reviews by three government agencies, most proposed power plant mergers over the past two decades have gained approval (Hempling 2018).

Firms cite various motives for acquisitions, including synergies, financial benefits, and complementarities between different asset types.¹⁰ Since fuel represents a large portion of operational costs, fuel efficiency improvements are often cited as a primary source of postacquisition cost savings.¹¹

C. *Electricity Production and Efficiency Measure*

A major challenge in analyzing merger efficiencies is the scarcity of suitable data, as most industries lack reliable cost and physical productivity measures (Asker and Nocke 2021). The US power generation industry is unusual in this respect because of the availability of high-frequency fuel efficiency data. This section describes the efficiency measures used in this study and explains the production process at power plants.

A power plant is an industrial facility that generates electricity. As of 2020, there were 11,070 utility-scale electric power plants in the United States (EIA 2020). Typically, power plants include multiple generators, transforming different forms of energy (primarily heat, wind, or solar) into electricity by using various production technologies. Our research focuses on fossil fuel generators, as their efficiency is more easily measured with available data.

Fossil fuel generators produce electricity by using the heat energy released from burning fuels (coal, natural gas, and oil).¹² In this process, the input is measured as the heat content of the fuel used in generation, while the output is measured as the electricity generated. This leads to a natural efficiency metric, called *heat rate*, which indicates how efficiently

⁹ In 2005, e.g., the Exelon-PSEG merger was not completed after failing to get approval from the New Jersey PUC (Wolak and McRae 2007; Morris and Oska 2008). In 2012, following the DOJ's request, Exelon Corporation and Constellation divested three plants in Maryland (Bushnell, Mansur, and Wolak 2012).

¹⁰ For many acquisitions in our sample, we accessed investor presentations and conference calls, which allowed us to identify the stated motives. Examples include (i) improvements in management (AES-DPL merger), (ii) cost synergies of \$175 million per year (NRG-GenOn merger), (iii) annual cost savings of \$150 million (Mirant-RRI Energy merger), and (iv) benefits of geographic, fuel, market, and earnings diversification (Vistra-Dynegy merger). Other motives include increasing the consumer base, diversifying the portfolio across technologies and regions, and accelerating efforts to comply with potential future environmental regulations.

¹¹ As an example, fig. OA-1 shows a slide from an investor presentation for the 2018 Dynegy and Vistra Energy merger, where firms claim that heat rate improvements will lead to \$30 million in cost savings.

¹² In thermal power plants, water is heated to generate steam, which moves through a turbine attached to a shaft. As the steam flows, it causes the shaft to spin, driving a generator that produces electricity.

a generator converts fuel into electricity. Heat rate is calculated as the ratio of the fuel's heat content (in million British thermal units; MMBtu) and the generator's electricity output (in megawatt-hours; MWh). Our measure of efficiency is the inverse of heat rate:

$$\text{Fuel Efficiency (Inv. Heat Rate)} = \frac{\text{Energy Output (MWh)}}{\text{Energy Input (MMBtu)}}. \quad (1)$$

Heat rate is the critical determinant of generator efficiency since fuel is the major input, representing 79% of operating costs.¹³ For this reason, it is a standard efficiency metric in the industry, commonly used by regulatory agencies and firms (EPRI 2014; EIA 2015).

Most importantly for this paper, fuel efficiency provides key advantages in analyzing the efficiency impacts of acquisitions. First, fuel efficiency is a quantity-based measure derived from input and output quantities rather than from revenues and input expenditures. Consequently, it is not directly affected by changes in input or output prices due to buyer and market power effects, allowing us to distinguish efficiencies from other merger-induced changes. Second, electricity is a homogeneous product, precluding any potential impacts on quality.¹⁴ Finally, the efficiency measure relies primarily on sensor data rather than survey responses, as is common in many other industries.¹⁵

Several factors can influence the heat rate in a power plant. Figure 1 displays a hypothetical example of a heat rate curve, where the line represents the heat rate and the bars represent a typical production distribution as a percentage of capacity. As suggested by the heat rate curve, a power plant's efficiency depends on its production level, typically reaching its peak when operating near full capacity. Moreover, fluctuations in production significantly affect efficiency. Given that electricity cannot be stored on a large scale and demand varies over time, power plants must frequently adjust their production in response to changing market conditions. These adjustments, known as *ramp costs*, reduce the overall efficiency of electricity generation (Borrero, Gowrisankaran, and Langer 2024).

Although electricity generation may appear relatively mechanical, the efficiency of generators in the United States shows notable variation, as

¹³ Based on the authors' calculations; see sec. B.5 of the appendix (available online) for methodology.

¹⁴ Some postacquisition changes, such as reliability and environmental performance, might be viewed as aspects of the "quality" of electricity generation. We will analyze these aspects later in the paper.

¹⁵ It is worth noting that our efficiency measure is fuel efficiency rather than total factor productivity and does not account for nonfuel inputs. While nonfuel inputs play a less significant role in electricity generation compared to other manufacturing industries, and substitution from fuel to other inputs is limited (Fabrizio, Rose, and Wolfram 2007), we explore them in sec. VI. We also provide a theoretical foundation of fuel efficiency based on a Leontief production function in sec. V.II.

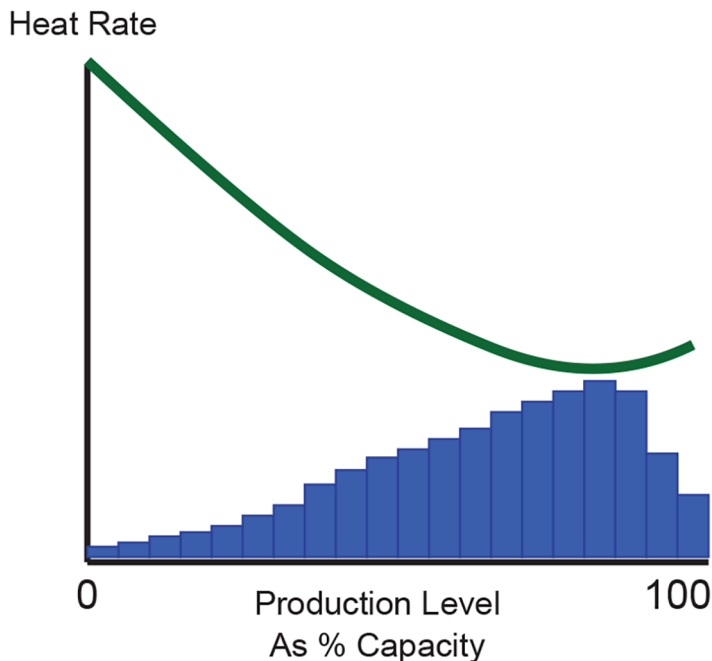


FIG. 1.—Representative heat rate curve in electricity generation, showing how heat rate changes with the production level. The bars represent a hypothetical distribution of production as a function of generator capacity.

in other manufacturing industries (Syverson 2011). Figure 2 shows the distribution of annual residual log productivity of generators after controlling for a rich set of observables, including ramp, generator age, fuel type, technology, capacity, boiler manufacturer, and model.¹⁶ The difference between the 90th and 10th percentiles of log productivity is 0.46, indicating that generators in the top decile are 58% more productive than those in the bottom decile.¹⁷ The large dispersion in residual productivity highlights the role of unobserved heterogeneity in efficiency and suggests that there is substantial room for efficiency improvements in many power plants.

¹⁶ We provide the details of this estimation procedure in sec. B.1 of the appendix.

¹⁷ The 90–10 percentile ratio of 1.58 is smaller than typical findings in other manufacturing sectors (Syverson 2011), likely because we condition on a richer set of observables than in other settings. Other researchers have also observed the heterogeneity in power plant productivity. Sargent and Lundy (2009), in a study commissioned by the Environmental Protection Agency (EPA), finds that the heat rates of coal-fired power plants range from 5 to 32.7 MMBtu/MWh. Staudt and Macedonia (2014) examine factors contributing to heat rate variation, including facility size, capacity factor, and coal type, and find considerable unexplained variability in heat rate.

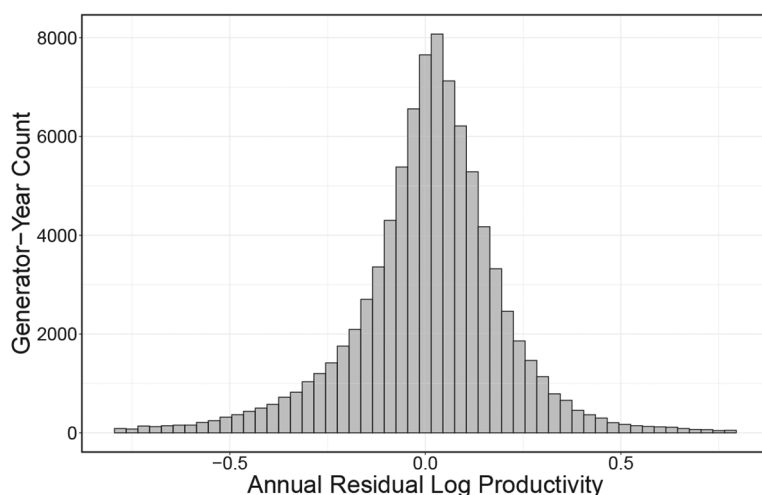


FIG. 2.—Distribution of residual yearly log productivity of fossil fuel generators in the United States between 2000 and 2023 after controlling for year, standard deviation of heat rate, generator age, fuel type, technology, capacity, boiler manufacturer, and model.

Improving the heat rate performance of a generator can be achieved in two main ways: (i) operational improvements and (ii) capital upgrades. Operational improvements, generally less costly than capital upgrades, include a range of practices such as installing control software, continuously monitoring unit and equipment performance, promptly repairing equipment impacting heat rate, training personnel, and implementing effective maintenance.¹⁸ Every year, power plant managers convene at the Electric Power Research Institute’s Heat Rate Improvement Conference to discuss these practices (EPRI 2022).¹⁹ The second approach to improving plant efficiency is to upgrade critical equipment, such as boilers, fuel feeders, and cooling systems, as old equipment deteriorates and new technology becomes available.

¹⁸ Several software products are available to monitor and improve power plant performance, e.g. PI Data Historian, EtaPRO/Virtual Plant, and Emerson Enterprise Data Server. Heat rates can also be improved with turbine enhancements, e.g., blade and seal repairs, cycle control optimization, boiler improvements, and deposit removal. Boiler improvements involve heat transfer surface maintenance, burner system inspection, and intelligent soot blower utilization. For methods of heat rate improvements and other examples, see EIA (2015), Emerson Process Management (2016), and Environmental Defense Fund (2017).

¹⁹ Figure OA-4 highlights a few case studies of heat rate improvements from the 2015 conference, as reported in Fitzgerald and Gelorme (2015). The following quote is particularly noteworthy: “For years we’ve talked about heat rate, but let’s be honest, in reality, it hasn’t driven maintenance and operational activities to a great degree” (2).

A critical factor influencing operational practices in a power plant is managerial and engineering input. As documented in detail in Bushnell and Wolfram (2009), the skills of key personnel can profoundly impact power plant performance. These personnel are responsible for continuously monitoring unit and equipment performance, conducting periodic tests to assess equipment condition, and planning production and maintenance schedules. Bushnell and Wolfram (2009) note the operator's impact as follows: "The act of balancing all of these input parameters was described by one manager as playing the piano and one star operator was considered a virtuoso on the instrument" (86).²⁰

Improving power plant efficiency is also crucial for environmental reasons. The higher a plant's efficiency is, the less fuel it requires, which leads directly to reduced emissions of local pollutants and greenhouse gases. As a result, enhancing fuel efficiency can be an effective method to mitigate emissions, a fact acknowledged by policymakers in the EPA's Clean Power Plan (EPA 2018).

III. Data and Summary Statistics

Our primary objective is to compile a dataset that allows us to construct a measure of generator efficiency and identify ownership changes. In this section, we outline our data sources and present summary statistics.

A. Data Sources

We combine data from FERC, the EPA, the Energy Information Administration (EIA), the North American Electric Reliability Corporation (NERC), S&P Global Market Intelligence (GMI, formerly SNL Financial), and Velocity Suite at the firm, plant, and generator levels for fossil fuel-fired power plants in the continental United States from January 2000 to March 2023. This section briefly describes the datasets, while section A of the appendix provides more detailed information on the data sources, variable construction, and descriptive statistics.

Generator- and plant-level data.—We use EIA, EPA, FERC Form 1, Velocity Suite, and GMI data to construct generator- and plant-level datasets. For generators, the information includes the installation year, fuel type, technology type, capacity, boiler model, and boiler manufacturer. For plants, we construct data on regulation status, location, ISO, and FERC region.

²⁰ As another example of the importance of personnel, PacifiCorp Energy states in its 2016 Heat Rate Improvement Plan that "continuous improvement and management of unit heat rates is the responsibility of all plant personnel" and that "good management of heat rate requires that plant management make optimizing heat rate a priority each day" (PacifiCorp Energy 2016, 14).

In addition, for approximately 35% of power plants, we have information on the number of employees, nonfuel costs, and capital expenditures.²¹

Production data.—We use the EPA's Continuous Emissions Monitoring Systems (CEMS) to obtain hourly input and output data. This dataset provides generation, heat input, and various emissions for nearly all fossil fuel units in the United States.²² Additionally, CEMS provides information on the environmental programs each generator is subject to and the scrubbers used for various pollutants. We merge this dataset with our generator- and plant-level data as detailed in appendix section A.1.

Ownership and acquisition data.—We construct a dataset on fossil fuel generator acquisitions by combining two separate datasets on ownership and transactions from GMI, as well as information from company press releases and newspaper articles.²³ The ownership data include all shareholders and their shares at both subsidiary and parent company levels. The transaction data provide details on transferred assets, transaction size, buyer and seller, announcement and closing dates, conference call transcripts, and descriptions. Given that virtually all power plant acquisitions in this industry require notification to regulatory agencies, this dataset provides comprehensive coverage of transactions during the study period.²⁴ It is well-known that ownership datasets may misidentify acquisitions by interpreting firm name changes and restructurings as ownership changes (Arora, Belenzon, and Sheer 2021; Davis et al. 2024). We address this issue by cross-referencing transaction and ownership data and by reviewing transaction descriptions, press releases, and news articles, as detailed in appendix section A.5.

Maintenance and outage data.—We obtain event-level data on outages, capacity reductions (derates), and maintenance from the Generating Availability Data System (GADS) database through a data-sharing agreement with NERC. This dataset covers all generators with a capacity over 20 MW, which are required to report events affecting their generation

²¹ The data sources for capital expenditures and nonfuel inputs are FERC Form 1 and Rural Utilities Service (RUS) Form 12, which are available only for major electric utilities as defined by FERC.

²² Every fossil fuel generator in the United States with a capacity greater than 25 MW must comply with the EPA CEMS program. This sample represents approximately 95% of the US fossil fuel-generating capacity.

²³ GMI, previously known as SNL Financial, collects data for the US electricity sector using regulatory filings from agencies like the Securities and Exchange Commission (SEC), FERC, RUS, EIA, and state-regulated utilities (GMI 2024). Additionally, it uses news aggregators to capture news articles, press releases, and corporate announcements. GMI has been widely used by researchers to study electricity markets (Davis and Hausman 2016; Jha 2020; Abito et al. 2022; Borenstein and Bushnell 2022).

²⁴ Before 2019, all power plant transactions required FERC approval regardless of size. On February 21, 2019, the FERC issued a rule setting a \$10 million threshold for approval and a \$1 million threshold for notification within 30 days. See <https://www.ferc.gov/news-events/news/ferc-issues-final-rules-revising-utility-merger-hydropower-regulations>.

capabilities to NERC. Available from 2013 to 2021, the data include each event's start and end times, type, and cause. The generator names are anonymized in this dataset, but information on capacity, state, fuel type, and monthly production hours is available. Using this information, we match this data to CEMS units using an algorithm described in appendix section A.7, achieving a match rate of 92.8% based on capacity.

Personnel data.—We compile monthly data on plant personnel from 2000 to 2020 using an EPA database of plant representatives, including names, tenure start and end dates, and contact information. We successfully matched about 70% of the personnel names to their LinkedIn profiles, thereby obtaining their title and education. Using LinkedIn data, we verified that 78% of the listed personnel are plant managers, while the remainder are primarily environmental compliance personnel and engineers. Thus, we consider plant representatives to be plant managers for the purposes of this study.

Other datasets.—We collect hourly data on ambient temperature and humidity from Velocity Suite for power plants in our sample, as weather conditions can affect generation performance. We also obtain firm-level industry classifications and the publicly listed status from GMI.

B. Construction of the Generator and Acquisition Sample

Our initial sample includes all electricity generators in the contiguous United States that operated between January 2000 and March 2023 and that are subject to CEMS regulations (5,876 generators). From this set, we exclude cogenerators that produce both steam and electricity, reducing the sample to 5,264 generators.

For acquisitions, we start with 5,216 generator acquisitions involved in any firm-to-firm transaction between January 2000 and March 2023. We eliminate acquisitions before a unit becomes operational or after its retirement (534) and minority acquisitions where less than 50% of the shares change ownership (864).²⁵ Next, we eliminate the ownership changes due to restructuring that happened mostly in the early 2000s because the source of efficiency improvements in these cases relates to incentive changes instead of acquisitions (Fabrizio, Rose, and Wolfram 2007; Cicala 2015). To do this, we primarily use data from Cicala (2015), Abito et al. (2024), and EIA electricity monthly reports. We further supplement these data with the regulatory status from EIA, Velocity Suite, and S&P Global and eliminate all ownership changes that result in a change in regulatory status within a 15-month window. Appendix section A.4 provides further

²⁵ A retired power plant may change ownership because of the value of its land or salvageable equipment or to transfer environmental cleanup responsibilities to the new owner.

details of this procedure. This step eliminates 615 acquisitions, reducing our sample to 3,515 generator acquisition events.

C. Descriptive Statistics on US Power Plant Acquisitions

This section presents descriptive statistics on fossil fuel power plant acquisitions. We demonstrate that the industry has undergone a substantial number of acquisitions, with significant heterogeneity in transaction, firm, and plant characteristics.

Figure 3A shows the share of fossil fuel electricity generation capacity that changed ownership between 2000 and 2023.²⁶ On average, 4.5% of the industry capacity changes ownership annually with some year-to-year fluctuations. As seen in figure 3B, these transactions vary widely in generation capacity. While most transactions include a few plants, there are some moderately sized transactions involving 5,000–10,000 MW capacity, as well as several large ones over 10,000 MW capacity.²⁷ This variation indicates that our evidence is not solely from a few large mergers, and we can test the heterogeneity of the effect by different transaction characteristics.²⁸

Table 1 presents summary statistics on generators, firms, and transaction characteristics. Our sample consists of 505 transactions, covering 3,515 distinct acquisition events that involve 2,048 unique generators. Most of these generators are gas fired (82%) and operate within an ISO (77%). About half of the acquisitions are cross-market transactions, where the acquirer does not own existing capacity in the acquisition market. The sample includes 244 unique acquirers and 224 unique target firms, with acquirers owning slightly more units than targets, on average.

In column 3, we present the same statistics but this time for the first acquisition of each generator, which forms our baseline sample in the empirical analysis. The observable unit characteristics are broadly similar between this subsample and all acquisitions (col. 2). Comparing acquired generators in column 3 with all generators in our sample (col. 1), we find no meaningful differences in capacity, average installation year,

²⁶ We define an acquisition as a change in ownership when a different firm gains the majority of the generator's shares after acquisition. In a small number of cases where no firm holds more than 50% of the shares, an acquisition is defined as a change in the largest shareholder.

²⁷ Table OA-1 (tables OA-1–OA-17 are in the online appendix) lists the 25 largest transactions during the sample period.

²⁸ Despite many acquisitions in the study period, there has been no significant change in market concentration in the United States as shown in fig. OA-5, which reports the national market shares of the largest five, 10, 20, and 30 firms by capacity owned. The concentration fluctuates but remains broadly stable because of significant entry and exit in the industry. Some examples can be seen in figs. OA-7 and OA-8, where we report firms with the largest capacity increase and decrease between 2010 and 2023.

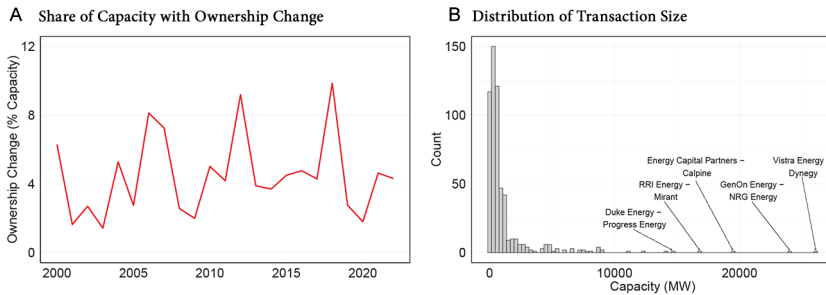


FIG. 3.—Summary statistics on power plant transactions. *A*, Annual percentage of fossil fuel electricity generation capacity that changed ownership in the United States from 2000 to 2023. *B*, Distribution of transaction sizes based on fossil fuel generation capacity in the United States during the same period; here, the unit of observation is a transaction, with the five largest transactions labeled.

and whether they operate in an organized market. However, we note differences in fuel type, with acquired generators more likely to be gas fired (77% vs. 71%). This trend primarily comes from (i) the large number of coal power plant retirements in the 2010s, (ii) a decline in acquisition opportunities due to the likelihood that coal plants are in regulated states, and (iii) uncertainty about the future of coal power plants (Davis et al. 2022). To address potential identification challenges arising from this and other potential differences, we control for monthly trends by fuel type, technology, capacity, and installation year in our empirical specifications.

Finally, the last two columns categorize the generators into the two acquisition types we identified: those involving both subsidiary- and parent-ownership changes and those involving only parent-ownership changes. Typically, a subsidiary is the legal entity that owns the power plant, while the parent company owns the subsidiary. Some transactions (asset acquisitions) involve changes in both subsidiary and parent ownership, whereas others (subsidiary acquisitions) involve changes only in parent ownership. Columns 4 and 5 list summary statistics for each generator's first acquisition of each type. We observe that these transaction types differ mainly in size, with parent-only ownership changes being significantly larger (an average of 15 vs. 5 units). This is consistent with the nature of parent-only ownership changes, which often involve taking over a large part of the target's portfolio.

We next document the firm composition in the industry. Figure 4 displays the evolution of ownership shares by the primary activity of the company (utilities, industrials, and financials) and by company type (publicly listed, private, or government owned). Figure 4A indicates an increasing presence of financial firms in the industry. The share of total capacity owned by financial firms rose from 2.5% in 2000 to 13.9% in 2023, suggesting a substantial reallocation of assets from utilities to financial firms.

TABLE 1
SUMMARY STATISTICS

	All Units (1)	All Acquisitions (2)	First Acquisitions (3)	Subsidiary/ Parent Change (4)	Only Parent Change (5)
A. Generator Characteristics					
Units	5,264	3,515	2,048	1,089	1,142
Plants	1,581	1,223	744	380	405
Unique units	5,264	2,048	2,048	1,089	1,142
Unique plants	1,581	726	726	373	400
Gas (%)	.71	.82	.77	.89	.68
Coal (%)	.18	.09	.12	.04	.17
Other fuels (%)	.10	.09	.11	.07	.15
Cross-market (%)		.57	.58	.51	.57
ISO (%)	.69	.77	.74	.81	.70
Unit capacity (avg.)	173.04 (184.75)	164.43 (159.01)	171.28 (173.39)	155.77 (145.02)	171.16 (179.41)
Installation year (avg.)	1,986.37 (20.08)	1,990.59 (16.24)	1,989.29 (17.27)	1,994.60 (14.25)	1,984.43 (17.86)
B. Firm Characteristics					
Acquirer firms		244.00	182.00	126.00	61.00
Target firms		224.00	159.00	111.00	70.00
Units acquirer owns (avg.)		45.81 (53.40)	45.94 (49.20)	50.72 (54.88)	39.35 (39.54)
Units target owns (avg.)		32.13 (47.23)	33.91 (49.57)	37.51 (50.64)	38.07 (53.30)
Acquirer firm capacity (avg.)		5,244 (8,698)	5,595 (9,112)	6,369 (9,605)	6,391 (9,507)
Target firm capacity (avg.)		7,314 (9,862)	7,466 (9,335)	8,312 (9,948)	6,200 (7,532)
C. Transaction Characteristics					
Transactions		505	318	213	72
Transaction size (avg. units)		7.0 (12.9)	6.4 (11.2)	5.1 (7.8)	15.9 (19.9)
Transaction size (avg. capacity)		1,191 (2,378)	1,164 (2,039)	812 (1,491)	2,909 (3,510)

NOTE.—This table includes summary statistics on acquisitions of fossil fuel-generating US units between 2000 and 2023. A description of the sample's construction can be found in sec. III.B. Each column reports the counts and characteristics of the data at varying sample restriction levels. Column 1 reports statistics from all generators in the data. Column 2 reports data from acquired generators. Column 3 restricts the acquisition sample to the first acquisition of each generator. Column 4 reports statistics for first acquisitions of each unit that involve both subsidiary- and parent-owner changes, while col. 5 focuses on the first parent-only owner change for each generator. Numbers in parentheses are standard deviations. The market definition for the cross-market calculations is the power control area. In cols. 2–5, the number of unique plants may differ from the total plant count, as in rare cases, units within the same plant were acquired at different times. Average acquirer and target characteristics report information before acquisitions. All capacity information is reported in MW.

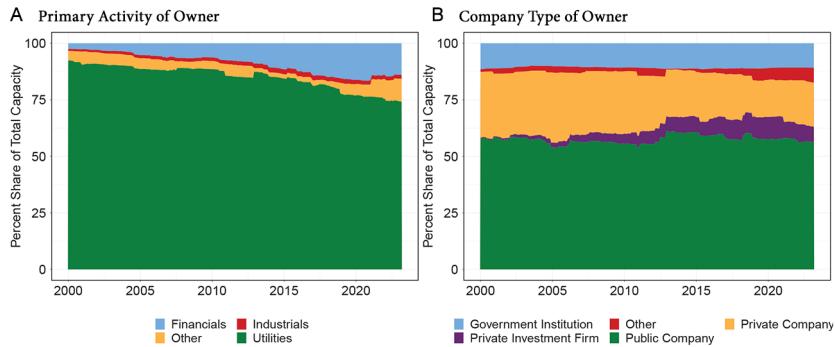


FIG. 4.—Share of generation capacity by firm type. *A*, Percent share of fossil fuel generation capacity in the United States between 2000 and 2023 by the primary industry type of the parent company. The “Utilities” category includes the generation capacity of both regulated and unregulated power plants owned by energy companies. *B*, Same statistics by categorizing parent owners into public companies, private investment firms, private companies, and government institutions. These financial firms primarily include private equity firms, pension funds, and bank funds. The classification is taken from GMI.

Figure 4*B* highlights that publicly listed firms own slightly more than half of the industry capacity, with their share remaining stable over time. Finally, government institutions—in rural areas, primarily local governments, with the exception of Tennessee Valley Authority—own 10.8% of total capacity.²⁹

IV. Empirical Results

Our empirical strategy aims to identify the effects of acquisitions on power plant productivity and other key operational outcomes. For this purpose, we implement a difference-in-differences research design that compares productivity trends of acquired generators to those that were never or not yet acquired. The main advantage of our empirical setting is the availability of a high-frequency measure of generator productivity, which enables us to track changes in productivity immediately before and after acquisitions.

We find that acquisitions increase the productivity of power plants by 2.0% on average, but with significant heterogeneity across acquisition

²⁹ An important institutional detail in electricity markets is variation in market structure and regulation across states. As shown in table 1, most acquisitions (77%) occur in organized markets where electricity prices are determined through competitive auctions. This trend is also reflected in the geographic distribution of acquisitions in fig. OA-9, where states with nonregulated wholesale generation have significantly higher numbers of acquisitions relative to their size (see also table OA-2). This highlights the potential role of regulatory institutions and market characteristics in shaping firms’ selection into mergers, which should be taken into account when interpreting our results.

types. In particular, ownership changes at both the subsidiary- and parent-owner levels lead to a 4.9% average efficiency gain, whereas ownership changes only at the parent-company level suggest no statistically significant efficiency effect. We conclude this section by examining the heterogeneity of the efficiency effect and studying how acquisitions affect other plant outcomes, such as generation and outages.

A. *Effects of Acquisitions on Efficiency*

We estimate the effects of acquisitions on efficiency using a regression of the following form:

$$y_{it} = \delta_1 \mathbf{1}_{\{\text{Before year 1}\}} + \delta_2 \mathbf{1}_{\{\text{After year 1}\}} + \delta_3 \mathbf{1}_{\{\text{After year 2}\}} + \delta_4 \mathbf{1}_{\{\text{After year 3}\}} + X_{it} + \alpha_i + \mu_t + \epsilon_{it}, \quad (2)$$

where y_{it} is the log efficiency of generator i at week t (measured as inverse heat rate) and α_i and μ_t are generator and week fixed effects, respectively.³⁰ The controls, X_{it} , in our preferred specification include ambient temperature and humidity, a dummy variable indicating whether the generator is subject to each environmental regulation, and pollution control device (scrubber) indicators for NO_x, SO₂, and particulate matter.³¹ Controlling for factors related to environmental regulations is important, as policy changes over the past two decades may influence firms' acquisition decisions or directly affect plant efficiency via scrubber installations.

In addition to these variables, we control for monthly time trends by state, installation year, fuel type, capacity bins, and technology type.³² By incorporating state-specific time trends, we account for changes in electricity demand and supply of non-fossil fuel generation. Furthermore, the time trends for generator characteristics allow for different efficiency trajectories based on generator type. For example, generators might experience a decline in efficiency over their lifespans, which can be nonlinear and vary by their vintage. We capture this variation by including installation year $\times$ month fixed effects.

³⁰ Even though the underlying data are hourly, we estimate this specification at the weekly frequency to alleviate computational complexity and reduce noise in the hourly data. We later perform a robustness check with daily frequency.

³¹ These programs are the Clean Air Interstate Rule NO_x Program, Nitrogen Oxides Budget Trading Program, Cross-State NO_x Program, Ozone Transport Commission Program, State Implementation Plan NO_x Program, Regional Greenhouse Gas Initiative, Clean Air Interstate Rule Ozone Season Program, Cross-State Ozone Season Program (Groups 1 and 2), New Hampshire NO_x Program, Mercury and Air Toxics Standards, Clean Air Interstate Rule SO₂ Program, Cross-State Ozone Season Program (Groups 1–3), Cross-State SO₂ Program, New Source Performance Standards for Toxics, and Texas SO₂ Program.

³² Capacity bins are categorized as 0–50, 50–100, 100–250, 250–500, and 500–2,000 MW; fuel types include gas, coal, and other; and technology types distinguish between combined cycle and other technologies.

The model in equation (2) includes coefficients of interest, δ_1 – δ_4 , to estimate efficiency effects from 1 year before acquisition to 3 years after acquisition.³³ We include δ_1 to examine potential preacquisition productivity effects that could arise because of anticipation effects or disruptions in the production process. The regression coefficients are normalized relative to the period 2 years before the acquisition, and standard errors are clustered at the acquisition level. We exclude acquired generators from the sample 3 years after their first acquisition to ensure that their post-treatment periods are not used as controls for other units. This means that we use only the first acquisition of each generator in our baseline empirical model, with a robustness check that includes all acquisitions presented in section VII.³⁴

It is worth noting that the unit of analysis is a generator rather than a plant. Although the same firm usually owns all the generators within a plant, generators often have distinct production profiles, maintenance schedules, and even retirement years (Gowrisankaran, Langer, and Zhang 2025). Therefore, we think the generator is the right level of analysis, and it is maintained throughout the paper unless otherwise stated.

Table 2 presents results with various sets of control variables (cols. 1–4) and different acquisition types based on subsidiary- and parent-owner changes (cols. 5 and 6). Our preferred specification with the full set of control variables in column 4 demonstrates efficiency increases following ownership changes. The efficiency of acquired generators increases by 0.6% 1 year after acquisition and reaches 2% after 2 years on average. The efficiency increase is robust to including a rich set of controls and time trends, and there is no efficiency change in the year leading up to the acquisition. Overall, these findings suggest that acquisitions lead to some improvements in generator efficiency.

Columns 5 and 6 of table 2 test whether the efficiency effect differs by the type of ownership change.³⁵ Column 5 shows the estimates only for acquisitions with ownership changes at both the parent and subsidiary

³³ Specifically, $\mathbf{1}_{\{\text{Before year 1}\}}$ is an indicator variable for 1–12 months before acquisition, $\mathbf{1}_{\{\text{After year 1}\}}$ for 0–12 months after acquisition, $\mathbf{1}_{\{\text{After year 2}\}}$ for 13–24 months after acquisition, and $\mathbf{1}_{\{\text{After year 3}\}}$ for 25–36 months after acquisition.

³⁴ One potential concern is that never-treated units operating in the same markets as treated units may be affected by acquisition through competitive spillovers, thereby violating the stable unit treatment value assumption (SUTVA). To address this concern, we implement a matching approach in sec. C.4 of the appendix that matches each treated unit only with those from different markets. Moreover, table OA-4 presents results based solely on small acquisitions—those involving less than 10% of market capacity and thus unlikely to generate significant spillovers. These estimates closely align with our main findings, suggesting that spillover effects, on average, are likely not substantial enough to impact our estimates.

³⁵ When estimating the effects of acquisition on one subsample of acquired units, we exclude the other acquired generators from the regression rather than grouping them with the never-acquired units so that they are not used as control units.

TABLE 2
EFFECTS OF ACQUISITIONS ON GENERATOR PRODUCTIVITY

	All Acquisition Types				Subsidiary and Parent Changes	Parent Changes Only
	(1)	(2)	(3)	(4)	(5)	(6)
Preacquisition (1 year)	.002 (.004)	0 (.003)	-.003 (.003)	-.003 (.003)	-.002 (.005)	-.007 (.003)
Postacquisition (1 year)	.018 (.006)	.015 (.006)	.006 (.005)	.006 (.005)	.015 (.007)	-.01 (.005)
Postacquisition (2 years)	.035 (.009)	.034 (.008)	.02 (.007)	.02 (.007)	.039 (.01)	-.003 (.007)
Postacquisition (3 years)	.039 (.011)	.036 (.01)	.02 (.009)	.02 (.009)	.049 (.012)	-.008 (.007)
Ambient temperature and humidity	X	X	X	X	X	X
Unit and week FE	X	X	X	X	X	X
State $\times$ month FE		X	X	X	X	X
Unit characteristic $\times$ month FE			X	X	X	X
Scrubber and environ- mental program FE				X	X	X
R^2	.707	.726	.752	.753	.763	.764
Observations (in millions)	1,838	1,838	1,838	1,838	1,494	1,575
Never-treated units	2,311	2,311	2,311	2,311	2,311	2,311
Treated units	2,046	2,046	2,046	2,046	1,089	1,142

NOTE.—This table presents the coefficient estimates of δ_1 , δ_2 , δ_3 , and δ_4 from estimating eq. (2). Columns 4–6 present our baseline specification, where we allow for time trends to vary flexibly by unit characteristic and include weather, scrubber, and environmental program controls. Unit characteristic fixed effects (FEs) include installation year, fuel, technology, and unit capacity bins. The dependent variable is the logarithm of efficiency, that is, the logarithm of the inverse heat rate. Standard errors are clustered at the acquisition level. Table OA-3 presents the same analysis results but for the subsample of acquisitions with both subsidiary- and parent-company changes.

levels. By contrast, column 6 includes ownership changes at only the parent level. The results suggest significant heterogeneity in the efficiency change based on acquisition type. When only the parent owner changes, the estimate is small and not statistically significant, whereas for both subsidiary- and parent-ownership changes, it indicates an efficiency increase of 4.9%. One might expect the efficiency effects to differ in these two cases because the subsidiary owners typically exert direct control over power plant operations and personnel, whereas the parent owners exercise indirect control through actions such as appointing directors, approving capital expenditures, and setting performance targets (Akey and Appel 2021). Furthermore, changes at the parent level are more likely to be financial acquisitions, potentially driven by motivations such as diversification and environmental policy considerations rather than efficiency gains

(Andonov and Rauh 2023). Overall, our results highlight that efficiency gains are influenced by the level of ownership change in the corporate structure and whether the direct owner changes.

After demonstrating the impact of acquisitions on generator efficiency, we shift our focus to the dynamic effects. Our goal is to determine the timing of efficiency changes and to test for different pretreatment trends between the acquired and other generators. To this end, we estimate the change in efficiency around the time of acquisition using the following specification:

$$y_{it} = \sum_{s \in (-36, 36)} \hat{\delta}_s D_{i(t+s)} + X_{it} + \alpha_i + \mu_t + \epsilon_{it}, \quad (3)$$

where $D_{i(t+s)}$ is a monthly indicator variable equal to 1 for generator i if it is acquired in month t and 0 otherwise. X_{it} includes the same control variables as before. Since we find efficiency effects only in acquisitions where both the subsidiary and parent owner change, we focus exclusively on those acquisitions hereafter.

The estimates of dynamic effects are shown in figure 5. The pre-acquisition coefficients are small and statistically insignificant, indicating similar productivity trends between acquired and nonacquired generators before acquisition. The difference between these groups remains small until 5 months after acquisition, at which point the efficiency of acquired plants begins to diverge. On average, the efficiency of acquired generators increases by 5% 18 months after acquisition and then stabilizes.

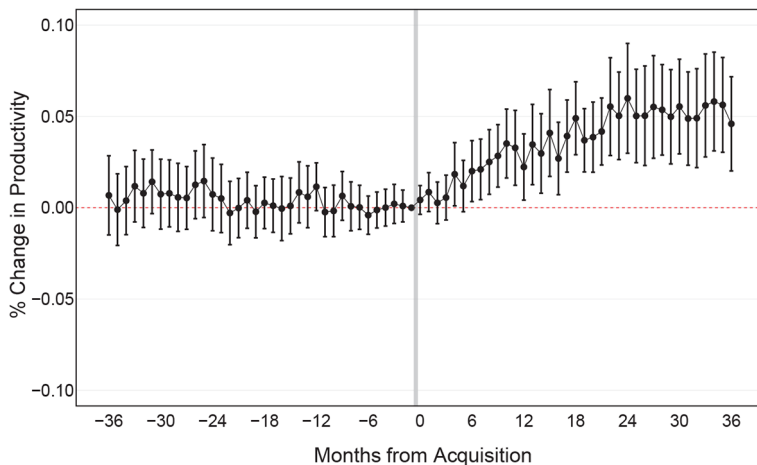


FIG. 5.—Dynamic effects of acquisitions on productivity. This graph presents the coefficient estimates of $\hat{\delta}_s$, where $s \in (-36, 36)$ from equation (3), along with 95% confidence intervals. The dependent variable is the logarithm of the inverse weekly heat rate. The unit of observation is generator-week. Standard errors are clustered at the acquisition level.

Not observing efficiency gains immediately after acquisitions suggests that the new owner requires time to implement efficiency improvements.

To interpret our results on efficiency gain as causal, we rely on the assumptions that an acquisition creates a discontinuous change in power plant behavior and that any unobservable efficiency trends that might lead to selection would be gradual enough to be distinguishable from the more discrete acquisition effect. Our data-rich setting offers key advantages for these assumptions to hold, as we observe production at short intervals and incorporate flexible time trends that account for factors likely to influence selection into acquisitions. Additionally, parallel trends holding 3 years before acquisition, coupled with the productivity increase beginning just a few months after acquisition, further suggest that efficiency gains are not likely caused by unobserved confounding factors.

Still, ownership changes are, of course, not random, and unobservable factors could influence efficiency without acquisitions. If acquirers observe these factors, that might lead to reverse causality, with acquisitions made in anticipation of efficiency gains. Although we cannot eliminate all potential identification threats or account for every unobservable factor, we conduct several robustness checks to ensure our results are robust to various specification choices and identification threats. For example, one possible scenario is that the acquirer observes that the target plant's manager will retire soon and decides to buy the plant, anticipating that the new manager will improve efficiency. To address such a concern, we estimate the effects of manager changes on efficiency in the absence of mergers and find no efficiency increase (fig. OA-10). In addition, we do a battery of robustness checks, including matching estimators, the Callaway and Sant'Anna (2021) estimator, estimation with daily data, estimation with net generation, weighted estimation, and placebo tests with minority acquisitions. We find that the results are robust to these specification choices.³⁶ See section VII for a summary of robustness checks and section F of the appendix for the corresponding results.

The results so far suggest that the efficiency of power plants improves following ownership changes. Yet it is important to recognize that efficiency gains in power plants can occur in various ways, not all of which are socially beneficial. For instance, generators might improve their average efficiency by decreasing production and reducing ramping, but this

³⁶ We emphasize that our estimates report the average treatment effect on the treated (ATT), specifically the efficiency effects of the proposed acquisitions. In our setting, the ATT, not the average treatment effect (ATE), is the primary and policy-relevant object of interest because we want to learn the effects of actual acquisitions, not hypothetical ones that would occur at random. However, this requires interpreting our results with an understanding of the circumstances under which mergers are proposed and of the existing regulatory approach to merger review. As a result, our results do not directly allow for a counterfactual analysis under specific changes in antitrust policy, e.g., those considered in Bhattacharya, Illanes, and Stillerman (2024).

could lead to increased production from a high-cost generator. Alternatively, new owners might operate generators more intensively, increasing their short-term efficiency but potentially causing increased outages and declining long-term performance. In the rest of this section, we provide additional analyses to gain insights into efficiency gains while reserving a more formal investigation of underlying mechanisms for the next section.

We examine the effect of ownership changes on various operational outcome measures, including generation, capacity utilization, operating hours, outages, and the carbon intensity of production. Capacity utilization is defined as the average hourly production as a proportion of capacity over a week, conditional on operation. Operating hours are calculated as the total hours a unit is operational in a given week. For outages, which are available between 2013 and 2021, we calculate the share of hours in a given week that a unit experiences a forced outage or derate. Finally, the CO₂ intensity is calculated by dividing CO₂ emissions by generation. Using these outcome measures, we estimate the same specification as in equation (2).³⁷

The coefficient estimates in table 3 indicate improvements in plant performance across multiple dimensions. In column 1, we find that acquired generators increase their generation by 7.3% compared to the baseline following the acquisition, so the efficiency improvements do not come at the expense of a decline in production.³⁸ Columns 2 and 3 suggest that generation increases at both the intensive and extensive margins. We observe that acquired power plants increase capacity utilization by 1.5 percentage points and operating hours by about 1 hour, though the latter is not statistically significant at the 5% level. The increase in utilization can be viewed as another source of productivity gain, as the generator produces more output conditional on the existing capital and labor stock, as argued by Braguinsky et al. (2015). Moving to column 4, the results indicate improvements in reliability, with a 6.3 percentage point reduction in average probability of forced outages and derates, suggesting that efficiency gains are achieved without compromising reliability.

³⁷ We estimate the first four outcomes in levels rather than logarithms to account for the presence of zeros. For generation, the distribution shown in fig. OA-6 indicates a thin upper tail, so a small number of high-output units is not likely to drive the results. Nevertheless, we estimate the same specification using a Poisson regression in table OA-8 and find a 4.1% increase in generation and a 2.2% increase in capacity utilization.

³⁸ This regression also offers indirect evidence that acquirers do not exert market power by withholding the output of the acquired generators. Moreover, this result does not necessarily imply that total market quantity increases. The capacity of acquired generators is typically small compared to the overall market, as illustrated in fig. OA-2. Because short-term electricity demand tends to be inelastic, any unit-level generation increase typically occurs because of changes in the unit's position on the dispatch curve—through either reduced downtime or a decrease in marginal cost—which reallocates generation from other units.

TABLE 3
EFFECTS OF ACQUISITIONS ON GENERATOR PERFORMANCE MEASURES

	Total Generation (MWh) (1)	Capacity Utilization (2)	Operating Hours (hours) (3)	Forced Outages/ Derates (4)	Log CO ₂ Inten- sity (tons per MWh) (5)
Preacquisition (1 year)	-169.221 (186.918)	.003 (.004)	-1.556 (.882)	-.003 (.012)	.006 (.006)
Postacquisition (1 year)	192.18 (267.302)	.006 (.005)	-.096 (1.137)	-.026 (.015)	-.009 (.007)
Postacquisition (2 years)	457.981 (346.979)	.013 (.005)	.584 (1.427)	-.034 (.019)	-.037 (.01)
Postacquisition (3 years)	527.331 (383.545)	.015 (.006)	.985 (1.593)	-.063 (.02)	-.046 (.013)
Ambient temperature and humidity	X	X	X	X	X
Unit and week FE	X	X	X	X	X
Unit characteristic × month FE	X	X	X	X	X
Scrubber and environ- mental program FE	X	X	X	X	X
Preacquisition mean	7,207.429	.665	45.116	.188	-.419
R ²	.797	.595	.695	.243	.842
Observations (in millions)	2.612	1.494	2.612	.705	1.418
Controls	2,311	2,311	2,311	1,383	2,026
Treated units	1,089	1,089	1,089	409	977

NOTE.—This table presents the coefficient estimates of δ_1 , δ_2 , δ_3 , and δ_4 from estimating eq. (2). Unit characteristic fixed effects include state, installation year, fuel type, technology type, and unit capacity bins. The unit of observation is generator-week. Standard errors are clustered at the acquisition level. The number of observations in col. 4 is lower than the rest because the outage and maintenance data begin in 2013. Some units in col. 5 are missing because their CO₂ emissions always equal zero in the data. The corresponding event study figure for each regression is reported in fig. OA-13. Columns 2 and 4 report unitless ratios between 0 and 1.

Finally, we note a 4.6% decrease in CO₂ intensity, mirroring the results on efficiency gains, as CO₂ emissions are inversely proportional to heat input.³⁹

B. Discussion of Results

Our findings in this section suggest that acquisitions lead to a 5% average increase in efficiency but only when both subsidiary and parent owners

³⁹ The effects on the other pollutants—SO₂ and NO_x—are also similar, with both falling by roughly 6%, as shown in table OA-9 and fig. OA-14. This suggests that efficiency improvements do not come at the expense of worsened environmental performance.

change. Additionally, acquired generators tend to increase production and utilization, reduce outages, and improve emission intensity. How large is the average 5% efficiency gain? To interpret this finding, it is helpful to compare our estimates to the average within-generator productivity growth in this industry, which is only 0.3% annually.⁴⁰ Given this modest within-generator productivity growth, the efficiency gains due to ownership changes are particularly noteworthy.

It is important to note that our analysis focuses on short-run to medium-run efficiency and performance, up to 3 years. Power plants are long-lived assets, and heat rate efficiencies may come at the cost of potentially increased mechanical stress or wear and tear over a horizon longer than 3 years. These dynamics suggest that heat rate efficiencies may not fully capture the long-term trade-offs related to productivity. Although the reduced precision of the estimates limits our ability to examine very long horizons, we report the 5-year postacquisition estimates for our main results in the appendix.⁴¹ The results clearly show a persistent improvement in heat rate over a 5-year horizon, whereas for other outcomes, the patterns are similar to our baseline results but estimated with less precision.

We also estimate the reduction in CO₂ emissions attributable to acquisitions. As detailed in appendix section B.3, our analysis assumes that efficiency gains begin after each unit's first acquisition and that their production levels remain unchanged after acquisition. Under these assumptions, we calculate a cumulative decrease of approximately 360 million tons in CO₂ emissions due to acquisitions from 2000 to 2023. This reduction is equivalent to the savings from replacing 800 TWh of gas-fired electricity generation with renewables.

C. What Predicts Efficiency Gains? Heterogeneity Analysis

This section explores whether efficiency gains are associated with observable plant or firm characteristics. While these findings do not establish causality, they help derive insights applicable to other industries by documenting transaction characteristics that could predict efficiency gains. For this estimation, we modify equation (2) by interacting treatment indicators with observable variables Z_{it} :

⁴⁰ Refer to fig. OA-11, which illustrates the average year-to-year within-generator productivity growth for generators not involved in acquisitions. The productivity growth fluctuates around zero, averaging a 0.3% annual increase over the sample period.

⁴¹ See table OA-11 and fig. OA-16 for heat rate estimates and fig. OA-17 for the other measures.

$$\begin{aligned}
y_{it} = & \delta_1 \mathbf{1}_{\{\text{Before year 1}\}} + \delta_2 \mathbf{1}_{\{\text{After year 1}\}} + \delta_3 \mathbf{1}_{\{\text{After year 2}\}} \\
& + \delta_4 \mathbf{1}_{\{\text{After year 3}\}} + \bar{\delta}_1 \mathbf{1}_{\{\text{Before year 1}\}} \times Z_{it} \\
& + \bar{\delta}_2 \mathbf{1}_{\{\text{After year 1}\}} \times Z_{it} + \bar{\delta}_3 \mathbf{1}_{\{\text{After year 2}\}} \times Z_{it} \\
& + \bar{\delta}_4 \mathbf{1}_{\{\text{After year 3}\}} \times Z_{it} + X_{it} + \alpha_i + \mu_t + \epsilon_{it}.
\end{aligned} \tag{4}$$

We estimate this equation separately for a set of generator, firm, or transaction characteristics that might be indicative of efficiency gains. In particular, we consider generator capacity, generator age, whether the acquirer is a serial acquirer, acquirer size, and whether the acquisition is a cross-market acquisition. Details on the construction of these variables are provided in section B of the appendix.⁴²

Results, reported in table 4, reveal that the efficiency increase is 3.4 percentage points larger when the generator capacity is higher than the median of acquired generator capacity. This suggests that acquirers might have stronger incentives to improve efficiency in larger plants, where the returns on such improvements are potentially higher. We do not find any significant differential effect with respect to generator age, as shown in column 2. Next, we turn to firm characteristics: whether the acquirer is a serial acquirer and acquirer size (total owned preacquisition fossil fuel generation capacity). The results, reported in columns 3 and 4, indicate that efficiency improvements are 5.8 percentage points higher when the acquirer is a serial acquirer and are 4.1 percentage points higher when the acquirer firm is larger than the median acquirer. These findings suggest that a firm's experience in plant operation and acquisitions could explain efficiency gains. They also align with and complement the findings of Hortaçsu et al. (2019) that large power firms are more sophisticated in bidding in wholesale electricity market auctions.⁴³ Finally, in column 5, we explore whether the efficiency effects differ for cross-market acquisitions. We categorize a generator acquisition as a cross-market acquisition if the acquirer owns no fossil fuel generation capacity in the acquisition market (defined as a power control area). We find that cross-market acquisitions exhibit 3.9 percentage point lower efficiency gains compared to within-market acquisitions.

We note that the heterogeneous effects observed in this analysis could arise from the types of acquisitions that select into each category or from

⁴² We also estimate the heterogeneous effects in a single regression that includes all interaction terms and report the estimates for the 10 most common acquisition types based on these observables in table OA-10.

⁴³ Hortaçsu et al. (2019) explore a counterfactual scenario in which large firms acquire smaller ones and improve their bidding operations. The evidence presented in this paper essentially validates the counterfactual hypothesis proposed by Hortaçsu et al. (2019). For further discussion, see sec. VI.B.

TABLE 4
HETEROGENEOUS EFFECTS OF ACQUISITIONS ON PRODUCTIVITY

	Capacity > Median (1)	Age > Median (2)	Serial Acquirers (3)	Firm Size > Median (4)	Cross- market Acquisitions (5)
Postacquisition (1 year) $\times$ Z	.023 (.011)	-.001 (.012)	.014 (.012)	.012 (.012)	.002 (.012)
Postacquisition (2 years) $\times$ Z	.035 (.015)	.004 (.016)	.059 (.016)	.049 (.017)	-.021 (.015)
Postacquisition (3 years) $\times$ Z	.034 (.018)	-.011 (.02)	.058 (.02)	.041 (.02)	-.039 (.019)
Ambient tem- perature and humidity	X	X	X	X	X
Unit and week FE	X	X	X	X	X
Unit character- istic $\times$ month FE	X	X	X	X	X
Scrubber and environmen- tal program FE	X	X	X	X	X
R^2	.763	.763	.763	.763	.763
Observations (in millions)	1,494	1,494	1,494	1,494	1,494
Units	2,311	2,311	2,311	2,311	2,311
Acquisitions	1,089	1,089	1,089	1,089	1,089

NOTE.—This table presents the coefficient estimates of $\tilde{\delta}_2$, $\tilde{\delta}_3$, and $\tilde{\delta}_4$ from estimating eq. (4). Each column reports results from a different regression by varying the interaction variable, Z. Unit characteristic fixed effects include state, installation year, fuel type, technology type, and unit capacity bins. The unit of observation is generator-week, and the dependent variable is the logarithm of the inverse weekly heat rate. Standard errors are clustered at the acquisition level. Appendix sec. B.4 provides details about the heterogeneity variables. See table OA-5 for the full set of estimates, including δ_1 – δ_4 and $\tilde{\delta}_1$ – $\tilde{\delta}_4$.

the inherent characteristics of those acquisition types. For example, the lower efficiency effects in cross-market acquisitions may occur because merging parties do not require strong efficiency gains to propose the merger, as regulators are less likely to challenge it given the absence of market power concerns. Alternatively, cross-market mergers might inherently lack the synergies or firm-specific specializations that typically occur within a single market, resulting in lower efficiency effects. Although our results in this section do not identify the exact mechanisms, the following section provides further insights into the specific sources of efficiency gains.⁴⁴

⁴⁴ Another important potential source of heterogeneity is market and state characteristics, as we showed that most acquisitions happen in organized markets and deregulated

V. Mechanisms

This section proposes mechanisms of efficiency gains, tests them empirically, and quantifies their role using a model of production in power plants. The key finding is that the majority of efficiency gains come from increasing productive efficiency within a generator.

A. Mechanisms of Efficiency Improvements

Two mechanisms could explain the estimated efficiency gains: (i) productive efficiency and (ii) dynamic efficiency. We first define these mechanisms and then develop a testable prediction for each one.

Productive efficiency.—Productive efficiency arises when the plant's new owner implements operational processes or invests in new equipment that improves efficiency. This mechanism occurs solely through increasing the generator's efficiency, enabling it to produce more with less fuel for a given production level. Therefore, it is independent of changes in the ramp profile or synergies with other plants in the same market. As illustrated in figure 6A, an implication of productive efficiency is a lower heat rate curve, leading to the following testable prediction:

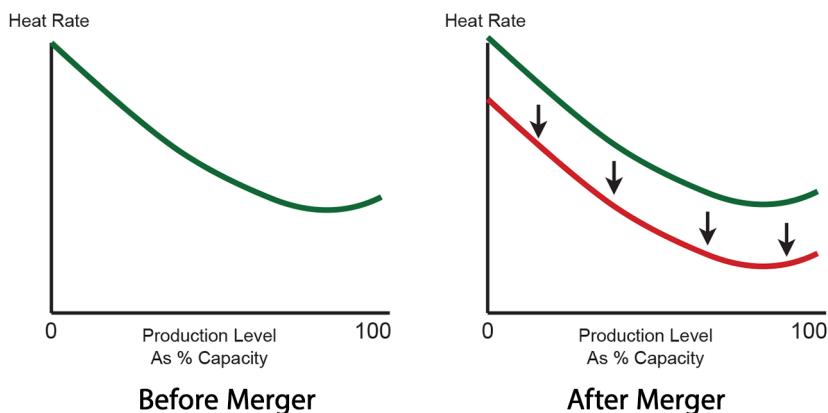
PREDICTION 1. If acquirers improve productive efficiency, the generator's heat rate curve shifts down.

Dynamic efficiency.—Dynamic efficiency arises from changes in the generation level over time. As discussed in section II.C, a key aspect of power generation is that efficiency is influenced by both the level of production and changes in production. Generators that experience significant production shifts incur ramp costs, which reduce overall efficiency. Power plants must manage these ramp costs because of the stochastic nature of electricity demand, which requires coordination between trading desk personnel responsible for submitting supply bids and plant operators overseeing production. Jha and Leslie (2023) note that uncertainty in residual demand or mismanagement in production can significantly increase ramp costs.⁴⁵ Figure 6B illustrates the dynamic efficiency effect, showing

states. To analyze this, table OA-6 examines heterogeneity based on three factors: (i) units operating in organized markets (ISO), (ii) units located in states with high acquisition activity (relative to the median), and (iii) units in deregulated states. We find that if a unit is in an ISO or in a deregulated state, the efficiency impact is about 4%–5% larger. Additionally, the efficiency impact is 1.8% larger in high-acquisition-activity states (though not significant at the 5% level). These results stress the importance of regulation and market structure in this industry, as markets and restructuring are associated with increased acquisition activity, which has a higher impact on efficiency.

⁴⁵ One source of dynamic efficiency could be portfolio effects, where a firm operating multiple power plants in the same market could achieve portfolio-level efficiencies through

A Productive Efficiency



B Dynamic Efficiency

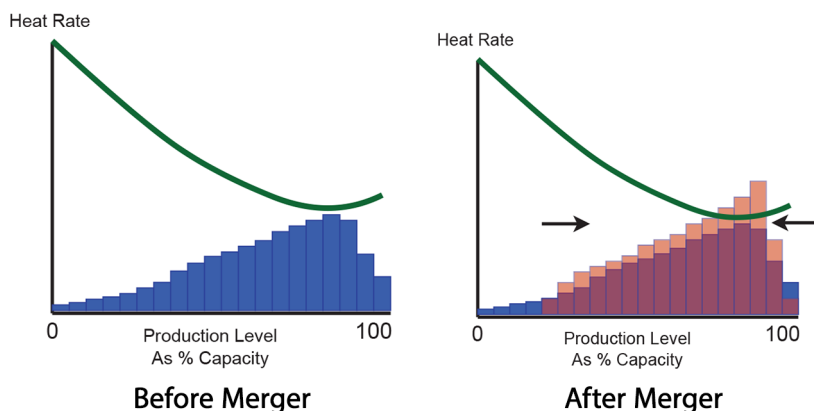


FIG. 6.—Illustration of mechanisms of efficiency gains studied in section V.A. *A*, Efficiency gains through productive efficiency, where the generator's heat rate curve shifts downward, producing greater efficiency at every capacity level. *B*, Dynamic-efficiency mechanism, where the heat rate curve remains unchanged but the production distribution becomes more concentrated because ramping declines.

a more concentrated production distribution and, therefore, lower ramp costs after acquisition. A testable hypothesis derived from this mechanism is the following:

ramp synchronization and efficient production allocation. Separating this mechanism from dynamic efficiency is challenging; therefore, we consider it part of dynamic efficiency effects.

PREDICTION 2. If acquirers improve dynamic efficiency, the variation of generation goes down.

B. Quantifying Productive Efficiency Using Production Functions

We start by testing for productive efficiency using an empirical strategy guided by prediction 1. In particular, we estimate a production function for generators, where we model heat rate with the following equation:

$$y_{it} = f_{it}(Q_{it}, R_{it}, X_{it}) + \epsilon_{it}, \quad (5)$$

where $y_{it} = \log(\text{Fuel}_{it}/Q_{it})$ is log heat rate, Q_{it} is production of generator i at hour t , and R_{it} is the ramp rate defined as the hourly change in production, $(Q_{it} - Q_{it-1})/C_i$, where C_i denotes generator capacity. The other variables, X_{it} , include ambient temperature and ambient humidity. Subscript i denotes the generator, t denotes the hour, and τ indicates the pre- or postacquisition period.

As described in Bushnell and Wolfram (2005), this form of heat rate modeling can be microfounded from a Leontief electricity production function. To see this, assume electricity is produced according to the following production function:

$$Q_{it} = \min(g_i(F_{it}, R_{it}, X_{it})\epsilon_{it}, h_i(K_{it}, L_{it})\omega_{it}), \quad (6)$$

where F_{it} , K_{it} , and L_{it} are fuel, capital, and labor inputs, respectively; and ϵ_{it} is unobserved, time-varying fuel efficiency; X_{it} are observable factors affecting fuel efficiency; and ω_{it} is total factor productivity. This Leontief production function, under a cost minimization assumption, implies that $Q_{it} = g_i(F_{it}, R_{it}, X_{it})\epsilon_{it}$. Assuming $g_i(\cdot)$ is strictly monotone in F_{it} , it can be inverted to write $F_{it} = g_i^{-1}(Q_{it}, R_{it}, X_{it})\epsilon_{it}$. Dividing both sides by Q_{it} and taking the logarithm yields the functional form in equation (5).

Importantly, the production function in equation (5) is indexed by i and τ , where τ equals 1 in the postacquisition periods and 0 in the pre-acquisition periods. Therefore, we estimate a generator-specific production function separately for the pre- and postacquisition periods, with f_0 representing the production technology of generator i before the acquisition and f_1 representing it afterward.

It is worth highlighting the benefits of estimating generator-specific production functions. The form in equation (5) accommodates heterogeneity in production technology across generators through the generator-specific

and time-varying production function f_{it} . Since f_{it} captures productivity differences across generators and over time, the production function literature generally interprets ϵ_{it} as an ex post shock (or measurement error) to output that is orthogonal to inputs. Thus, our model is likely to be robust to transmission bias, which creates a correlation between productivity level and inputs (Marschak and Andrews 1944; Akerberg, Caves, and Frazer 2015). Furthermore, through a time-varying production function, we model the effects of acquisitions not only on the productivity level but also on the production technology.

We can estimate a flexible production model because of the availability of hourly data, as it provides a large number of observations for each generator, even within a limited time frame around acquisitions. This highlights the advantages of a data-rich environment, contrasting with the production function literature, which often imposes an industry-level functional form because of data limitations (De Loecker and Syverson 2021).

We use a nonparametric local polynomial regression to estimate the functions f_0 and f_1 for each acquired generator as detailed in appendix section B.2. To estimate f_1 , we use 3 years of postacquisition data, while f_0 is estimated using data from 3 years prior to the acquisition. We then measure the changes in productivity by calculating the difference between the post- and preacquisition heat rate curves for each generator and then averaging these differences. Specifically, we calculate

$$\Delta C(Q) = c_{\text{post}}(Q) - c_{\text{pre}}(Q) = \frac{1}{N_{\text{acq}}} \sum_{i=1}^{N_{\text{acq}}} (f_1(Q, \bar{R}_i, \bar{X}_i) - f_0(Q, \bar{R}_i, \bar{X}_i)),$$

where N_{acq} represents the number of acquired generators and $Q \in [10, 100]$ is the production level as a percentage of capacity.⁴⁶ The terms $c_{\text{pre}}(Q)$ and $c_{\text{post}}(Q)$ denote the average heat rate at production level Q before and after acquisition, respectively. The control variables are set to $\bar{R}_i$ and $\bar{X}_i$, which is 0 for ramp rate, and the preacquisition medians for temperature and humidity to isolate the effects of postacquisition changes in these variables. Thus, $\Delta C(Q)$, known as the average structural function (Blundell and Powell 2003), represents the change in the average heat rate at each production level controlling for ramp and weather conditions.

We also construct a control group by matching each acquired generator to those never acquired in a different market based on capacity, age, fuel, and technology type, as detailed in appendix section B.2. We then apply

⁴⁶ The utilization values start at 10% because production at lower capacity levels is rare and tends to yield noisy estimates.

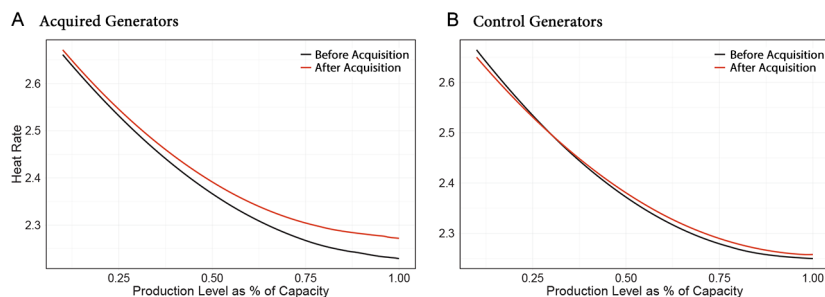


FIG. 7.—Estimates of average heat rate curves 3 years before acquisition and 3 years after acquisition for the acquired generators group (A) and the control group (B) constructed by a matching procedure detailed in section C.4 of the appendix. The treated group sample is the same as column 5 of table 2. Figure OA-12 reports the confidence band for the difference between the two heat rate curves obtained from a bootstrap procedure. Further details of the estimation procedure are provided in appendix section B.2.

the same estimation procedure to these control generators to quantify changes in the heat rate curves without acquisitions.

Figure 7A reports $c_{\text{post}}(Q)$ and $c_{\text{pre}}(Q)$ for the acquired generators, while figure 7B displays these curves for the control group. Comparing pre- and postacquisition heat rate curves reveals a downward shift in the heat rate curve for acquired generators at every production level, with larger effects near the generator's capacity. In contrast, the control group's heat rate curve remains stable.⁴⁷ We also calculate a confidence band for the difference between the pre- and postacquisition heat rate curves of acquired generators, as presented in figure OA-12, confirming that the difference is statistically significant. These results provide direct evidence that the acquirers increase the productive efficiency of the acquired generators by improving their heat rates.

Having estimated the heat rate curves, we can now quantify the total efficiency gain from the downward shift in the heat rate curve. To do this, we integrate the difference between the post- and preacquisition curves as follows:

$$\Delta = \frac{1}{N_{\text{acq}}} \sum_{i=1}^{N_{\text{acq}}} \int (f_{i1}(Q, \bar{R}_i, \bar{X}_i) - f_{i0}(Q, \bar{R}_i, \bar{X}_i)) dF_{i0}(Q),$$

where $F_{i0}(Q)$ represents the preacquisition production distribution of generator i . This calculation maintains the production distribution from the preacquisition period and quantifies efficiency gains solely from

⁴⁷ The slight shift in the heat rate curve of control generators is consistent with the within-generator aggregate efficiency growth documented in fig. OA-11.

changes in the heat rate curve. The result indicates a 3.9% (confidence interval: 2.9%–4.8%) increase in efficiency, accounting for approximately three-quarters of the total efficiency gain observed in the event study. Therefore, most of the efficiency gain stems from increased productive efficiency attributable to the acquirers' improvements to the generator's internal operations.

C. *Quantifying the Dynamic Efficiency Mechanism*

We next assess the role of dynamic efficiency. Prediction 2 posits that increased dynamic efficiency results in reduced production variability after acquisition. To test this, we consider three measures of production variability: the CV of heat rate, the CV of utilization, and the number of ramps.⁴⁸ These metrics collectively provide insights into how acquisitions influence the production dynamics of a generator.

We estimate our baseline regression using these measures as outcome variables and report the estimates in table 5. After acquisition, we observe significant reductions in all measures of production variability. Specifically, the CV of heat rate decreases by an average of 0.029 from a preacquisition mean of 0.235, and the CV of utilization drops by 0.029 from a preacquisition mean of 0.364. We also find a significant decline in the number of ramps, showing a 12% decrease from the preacquisition level.

We can also quantify the contribution of the dynamic efficiency effect using the production model developed in section V.B as follows:

$$\frac{1}{N_{\text{acq}}} \sum_{i=1}^{N_{\text{acq}}} \left(\int f_{i1}(Q, R, \bar{X}_i) dF_{i1}(Q, R) - \int f_{i1}(Q, R, \bar{X}_i) dF_{i0}(Q, R) \right),$$

where $F_{i1}(Q, R)$ and $F_{i0}(Q, R)$ denote the distributions of production and ramp rate post- and preacquisition. This calculation essentially poses the following counterfactual question: What would be the efficiency difference if the generator had the postacquisition heat rate curve (f_{i1}) in both the pre- and postacquisition periods while changing only the production profile from f_{i0} to f_{i1} ? In other words, it controls for production technology and captures only those efficiency effects due to changes in the production distribution. We caution that this calculation reflects both changes in production variability and the effect of shifting output toward the plant's efficient scale, even though the latter is not strictly a dynamic efficiency. This

⁴⁸ We define a ramp event as a change in production where the output increases from below 20% to above 80% of the plant's capacity or decreases from above 80% to below 20% within a period of less than 3 days.

TABLE 5
REGRESSION RESULTS ON DYNAMIC EFFICIENCY MECHANISM

	CV of Heat Rate (1)	CV of Utilization (2)	Number of Ramps (3)
Preacquisition (1 year)	-.001 (.005)	0 (.004)	-.059 (.1)
Postacquisition (1 year)	-.016 (.007)	-.015 (.005)	-.265 (.138)
Postacquisition (2 years)	-.026 (.008)	-.026 (.008)	-.369 (.159)
Postacquisition (3 years)	-.029 (.009)	-.029 (.008)	-.435 (.167)
Ambient temperature and humidity	X	X	X
Unit and week FE	X	X	X
Unit characteristic $\times$ month FE	X	X	X
Scrubber and environmental program FE	X	X	X
Preacquisition mean	.235	.364	3.499
R^2	.195	.528	.452
Observations (in millions)	1.476	1.476	1.476
Never-treated units	2,309	2,309	2,309
Treated units	1,089	1,089	1,089

NOTE.—This table presents coefficient estimates of δ_1 , δ_2 , δ_3 , and δ_4 in eq. (2) from a regression of the CV of heat rate, CV of utilization, and number of ramps on treatment dummies. The coefficients are calculated from hourly data every week; thus, the regressions use weekly data. Unit characteristic fixed effects include state, installation year, fuel type, technology type, and unit capacity bins. Figure OA-15 reports each regression's corresponding event study figure. Standard errors are clustered at the acquisition level.

calculation yields an efficiency gain of 1.7% (confidence interval: 0.0%–4.2%), corresponding to around 30% of the total effect.⁴⁹

It is important to recognize that improvements in dynamic efficiency can arise from various factors. One potential factor is increased productive efficiency: a marginal generator in the dispatch curve that becomes more efficient after acquisition may operate inframarginally more often, leading to reduced ramping. Another explanation could be decreased outages and forced maintenance, which would reduce ramping between inactive and operational modes. Furthermore, the acquirer may change the power plant's operations or improve coordination between the bidding desk and plant operators. Although our analysis does not separate the impact of these individual sources, it highlights the importance of ramping costs in improving power plant efficiency.

The analysis in this section focused only on marginal cost gains from fuel efficiency. Acquisitions may also reduce fixed costs or result in nonfuel cost

⁴⁹ The productive and dynamic efficiency effects do not sum to exactly 100% because of noise in estimation.

savings. For example, decreased ramping can reduce wear and tear, thus lowering maintenance expenses and prolonging the lifespan of capital. Additionally, acquisitions may generate economies of scale in maintenance and bidding (Haldi and Whitcomb 1967; Hortaçsu and Puller 2008). Although these fixed cost efficiencies could be large, they are generally not considered in merger analysis (Röller, Stennek, and Verboven 2006) and cannot be accurately measured with our data. Therefore, they fall outside the scope of this paper.

VI. How Do Acquirers Improve Productive Efficiency?

So far, our analysis has demonstrated efficiency improvements following ownership changes, mainly due to increased productive efficiency. This result raises a natural follow-up question: How do acquirers achieve these efficiency gains? We will now address this question.

A. *Productive Efficiency: Operational Improvements or Investment?*

In section II.C, we proposed two potential mechanisms to improve a power plant's productive efficiency. The first mechanism involved implementing low-cost operational improvements, such as personnel training, efficient production management, best practices, and improvements in repairs and maintenance. Such improvements would indicate a knowledge transfer from the acquirer to the acquired plant. The second mechanism entailed high-cost capital investments by acquirers to upgrade existing equipment, suggesting that the previous owner faced credit constraints or lacked the incentives to make efficiency-improving capital investments.

Disentangling these two sources is useful not only for understanding the nature of efficiency gains but also for informing antitrust policy. To be recognized in merger evaluations, efficiencies must be specific to a merger.⁵⁰ Efficiencies from relaxing capital constraints would not qualify, as they could also be attained through alternative means, such as by raising new capital or minority investment. In contrast, knowledge transfers can be considered merger-specific efficiencies because they involve exchanging organizational knowledge and intangible capital between the merging entities, a process that is unlikely to occur outside of a merger (Atalay, Hortaçsu, and Syverson 2014).

We aim to disentangle the sources of productive efficiency improvements using additional data on manager changes, capital investments, nonfuel

⁵⁰ The 2023 Horizontal Merger Guidelines state, "the merger will produce substantial competitive benefits that could not be achieved without the merger under review" (DOJ and FTC 2023, 32).

inputs, and maintenance. Specifically, we investigate whether power plants undergo personnel changes and increase capital expenditures after acquisition. The former would suggest operational changes, whereas the latter would provide evidence for the role of capital investment. Moreover, by analyzing nonfuel inputs and maintenance, we evaluate the possibility of substituting fuel with other inputs to achieve efficiency gains.

We use the dynamic difference-in-differences specification in equation (3) to explore whether acquired plants experience more managerial changes than nonacquired plants. The dependent variable is set to 1 if the power plant manager is replaced in a given month and 0 otherwise. Results shown in figure 8 reveal a significant increase in managerial changes after acquisition: acquired plants are 15 percentage points more likely to experience a change within 1 month and 45 percentage points more likely within 2 months relative to the nonacquired plants. Using LinkedIn data, we also analyze the qualifications of new managers and find that new managers are 5.58 percentage points (standard error = 2.51) more likely to hold a master's degree and 4.26 percentage points (standard error = 1.79) more likely to have a bachelor's degree than those involved in changes without mergers.

The results on manager changes raise an important question: Can the efficiency gains be solely attributed to manager changes? To explore this, we estimate the efficiency effects separately for acquisitions with and without manager changes (reported in table OA-7) and for manager changes without acquisitions (reported in fig. OA-10). The findings indicate that

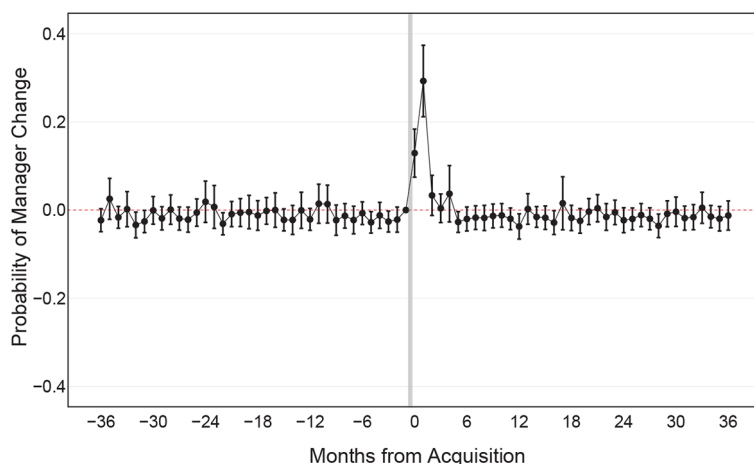


FIG. 8.—Effects of acquisitions on manager change. This graph shows coefficient estimates of a manager change dummy on δ_s , where $s \in (-36, 36)$. The unit of observation is generator-week. Error bars show 95% confidence intervals. Standard errors are clustered at the plant level.

manager changes without acquisitions have no significant efficiency effects, whereas acquisitions without manager changes still lead to large efficiency improvements. Although these results are not conclusive on their own, they suggest that manager changes alone are not sufficient to generate efficiency gains and that firms cannot achieve efficiencies simply by replacing their manager. Our interpretation is that manager turnover indicates significant operational changes, and it must be accommodated by organizational changes to generate efficiency gains. These insights echo the common findings in the literature on the role of management practices and organization in explaining productivity differences (Bloom and Van Reenen 2010; Macchiavello and Morjaria 2022).⁵¹

Next, we examine the changes in capital expenditures and nonfuel inputs after acquisitions, acknowledging that this analysis relies on a different and more limited dataset. Specifically, data on capital expenditures, number of employees, and nonfuel intermediate input costs are available for only a subset of plants reporting to FERC, and they are annual, unlike the hourly heat rate data. Therefore, while these findings provide useful insights, they warrant cautious interpretation given these data limitations.

The coefficient estimates in table 6 suggest that acquired plants do not increase capital expenditures. The coefficient estimate for capital expenditures is -24% , but it is imprecise because of the small sample size. Nevertheless, it is still possible to reject the hypothesis that capital expenditure increases by more than 5% at the 10% significance level.⁵² The estimates for nonfuel materials costs and labor in columns 2 and 3 are also noisy, but they similarly provide evidence against large increases. These findings suggest that efficiency increases do not come from capital expenditures; instead, operational improvements are the key drivers of increases in productive efficiency.

In our final analysis, we examine how maintenance changes after acquisition, as it could also be viewed as an input in electricity generation. Moreover, maintenance is important in its own right to understand plant performance because decreased forced maintenance might indicate better

⁵¹ To further understand the role of managerial changes in efficiency gains, table OA-12 reports the additional increase in the probability of a manager change across different types of acquisitions, using the observable characteristics from the heterogeneity analysis in sec. IV.C. This analysis suggests that the probability of a managerial change tends to be positively correlated with the likelihood of observing higher efficiency gains, providing suggestive evidence for the interpretation that managerial changes signal operational changes in the plant.

⁵² Further evidence against the capital expenditure hypothesis comes from the timing of efficiency gains and operating hours. Significant capital investments typically require more than 5 months to implement and usually involve considerable downtime, neither of which we observe.

TABLE 6
EFFECTS OF ACQUISITIONS ON NONFUEL COSTS AND MAINTENANCE

	Log Capital Expenditures (1)	Log Nonfuel Costs (2)	Log Employ- ees (no.) (3)	Maintenance Probability (4)
Preacquisition (1 year)	-.214 (.161)	-.335 (.4)	-.22 (.111)	-.013 (.009)
Postacquisition (1 year)	-.052 (.163)	-.112 (.211)	-.326 (.124)	-.024 (.012)
Postacquisition (2 years)	-.236 (.17)	.095 (.267)	-.06 (.136)	-.038 (.014)
Postacquisition (3 years)	-.236 (.176)	-.304 (.297)	-.003 (.156)	-.048 (.014)
Ambient temperature and humidity	X	X	X	X
Unit and week FE	X	X	X	X
Unit characteristic $\times$ month FE	X	X	X	X
Scrubber and environmental program FE	X	X	X	X
Preacquisition mean				.096
R^2	.896	.704	.946	.15
Observations (in millions)	.018	.018	.017	.705
Controls	1,472	1,643	1,553	1,383
Treated	176	203	148	409

NOTE.—This table presents the coefficients derived from estimating the effects of acquisitions on capital expenditures, nonfuel intermediate input costs, number of employees (all observed at the annual frequency), and maintenance (observed at the weekly frequency). Standard errors are clustered at the acquisition level. Note that the capital expenditure information is available only for major electric utilities as defined by FERC.

equipment management by new owners, which would increase production, as the generator would go offline less often for maintenance. We analyze the probability that a generator undergoes maintenance in a given week. The results in column 4 suggest that maintenance probability decreases after acquisitions, indicating that more maintenance duration is not the primary means of improving efficiency.

B. Who Acquires Whom? Productivity of Acquirer and Target Firms

This section estimates the productivity levels of acquirer and target firms to determine whether (i) acquirers are more productive than target firms and (ii) acquirers have a comparative advantage in utilizing acquired assets. This analysis not only provides evidence on the mechanisms of efficiency improvements but also offers insights into the broader economic implications of ownership changes. Acquisitions, as a key mechanism of resource reallocation among firms, can lead to allocative efficiency gains in the economy by transferring assets from less productive to more productive firms or enabling better utilization of these assets.

We modify our baseline specification in equation (2) by including three sets of indicator variables to estimate the efficiency levels of three distinct asset types: (i) acquired generators, (ii) the acquirer's existing generators not involved in the transaction, and (iii) the target's existing generators not involved in the transaction. Formally, we estimate the following specification:

$$y_{it} = \sum_{j=1}^3 \theta_{1j} \mathbf{1}_{\{\text{Before years } 1-3\}j} + \theta_{2j} \mathbf{1}_{\{\text{After years } 1-3\}j} + X_{it} + \mu_i + \epsilon_{it}, \quad (7)$$

where j represents the asset types listed above and y_{it} is log productivity.⁵³ This specification estimates the efficiency of the target's assets, acquirer's assets, and acquired assets around the time of acquisition. Note that this regression does not include generator fixed effects, as we aim to estimate level differences in productivity rather than changes. However, we account for generator characteristics by controlling for generator age, capacity, technology, fuel type, and scrubbers. We restrict the sample to transactions where both the acquirer and the target own generators not involved in the transaction.⁵⁴ We normalize the efficiency of the acquirer's generators to zero in the preacquisition period.

Figure 9 presents the estimated coefficients for three groups: the acquirer's existing assets, the target's existing assets, and the acquired assets. First, we observe that acquisitions do not significantly impact the productivity of existing assets, with the productivity levels of both acquirer and target remaining stable around the time of acquisition. Notably, however, acquirers have a productivity level 1.7% higher than target firms. As a result, acquisitions reallocate assets from less to more productive firms, although the difference in productivity is relatively modest.

Next, we compare the productivity of acquired plants with the acquirer's and target's existing plants. A key observation is that target firms tend to sell their underperforming assets: the sold plants are 3% less productive than other plants in the target's portfolio. What happens to these underperforming plants after acquisition? The efficiency of these plants improves by 5%, reaching the same efficiency level as the acquirer's other plants.

The findings in this section indicate that high-productivity firms buy underperforming assets of low-productivity firms and make the acquired assets as productive as their existing assets. This pattern corroborates our earlier conclusion that efficiency improvements come primarily from

⁵³ For the acquired plants, we exclude the first year after acquisition to estimate the long-term effects of acquisitions.

⁵⁴ This subset accounts for 67% of all acquisitions.

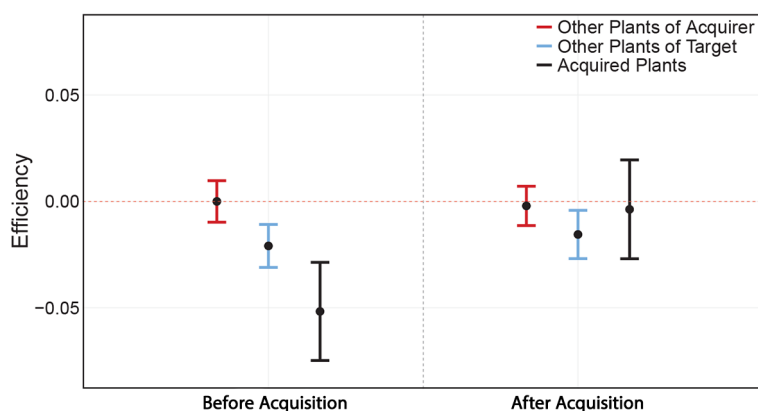


FIG. 9.—Efficiency of acquirer and target firms; regression estimates from equation (7). For both preacquisition and postacquisition periods, the bars represent the normalized efficiency of the acquirer's existing assets, the target's assets, and the acquired assets from left to right. Error bars indicate 95% confidence intervals. The efficiency of the acquirer's assets in the preacquisition periods is normalized to zero. Standard errors are clustered at the plant level.

operational improvements through knowledge transfers. Furthermore, these results also provide empirical evidence about the theories of merger gains in the literature. One common theory, the Q theory of mergers (Jovanovic and Rousseau 2002), posits that there are inherent productivity differences between firms and that acquisitions transfer assets from low- to high-productivity firms. This implies a high-buys-low pattern. According to another theory proposed by Rhodes-Kropf and Robinson (2008), assets and firms could be complementary, with firms having varying degrees of capability in operating different assets. This implies a like-buys-like pattern. Our results lend support to both theories of mergers by demonstrating that assets are allocated to firms with relative and absolute advantages in utilizing them.

This analysis also serves as an important input for merger analysis, particularly in determining postmerger marginal costs for firms with different efficiencies (Farrell and Shapiro 1990). A key question in this context is the transferability of efficiency between firms, as prior research has highlighted that organizational challenges in integrating firms can hinder the transfer of productivity-improving practices (Weber and Camerer 2003; Malmendier, Moretti, and Peters 2018). Our empirical analysis contributes to this question by providing evidence that efficiency could be transferable in the context of power plant acquisitions.

A natural question arising from this section's findings and the paper's overall conclusions is why previous owners do not implement the operational improvements. Given that our study is an industry-level analysis

rather than a firm-level case study, we cannot provide a definitive answer to this question. Nonetheless, it is important to note that our results align with substantial evidence of persistent firm-level productivity differences in various industries (Syverson 2011; Gibbons and Henderson 2012). We interpret our evidence to suggest that some firms develop intangible capital over time for more efficient power plant operation and that this within-organization knowledge is transferable through ownership changes. This can occur in many forms, for example, by transferring asset-specific expertise (Hortaçsu and Syverson 2007; Atalay, Hortaçsu, and Syverson 2014), operational strategies (Eliason et al. 2020), or managerial practices (Bloom, Sadun, and Van Reenen 2012). Therefore, acquisitions provide a channel for spreading intangible capital across firms, which is less likely to be achieved through other means.⁵⁵

VII. Robustness Checks

In this section, we explore the robustness of our findings by considering alternative specifications. Detailed descriptions of this analysis and the corresponding results are provided in appendix section C and appendix section F.

Estimation frequency.—Our main analysis uses weekly data to estimate the effects of acquisitions, as this aggregation reduces noise in the hourly data and is computationally convenient. To assess the robustness of our findings, we conduct the same estimation using daily frequency. The results, reported in figure OA-18, remain consistent at the daily frequency, although there is a slight increase in standard errors.

Acquisition sample.—In our baseline specification, we focus only on each generator's first acquisition to avoid using data from postacquisition periods. As a robustness check, we extend our analysis to include all acquisitions of generators during the sample period. The findings, reported in column 4 of table OA-13, figure OA-20, and table OA-16, suggest a slightly smaller effect than our baseline result, indicating that the efficiency gains may be lower with subsequent acquisitions.

Weighting by capacity.—Our main specification estimates the average effects without accounting for the varying capacity sizes of acquired generators. In a robustness check, we weigh observations by capacity, which provides a more accurate measure of total cost savings. The results from this

⁵⁵ This source of efficiency gains differs from the technology-related synergies studied in the merger literature. Some examples include economies of density in the ride-hailing industry (Rosaia 2020), congestion-related efficiency in the telecommunications industry (Elliott et al. 2025), and reduction in shipping distance in the beer industry (Miller and Weinberg 2017).

specification suggest similar efficiency effects, indicating that the evidence does not primarily come from small units (col. 2 of table OA-13, table OA-15, and fig. OA-23).

Estimation with net generation.—While our primary analysis uses gross generation because of its high-frequency availability, we also conduct a robustness check using net generation data from EIA. The results, reported in figure OA-22, column 3 of table OA-13, and table OA-17, are broadly similar to our main findings, though the effect is slightly lower.

Estimation after 2010.—A potential concern in our analysis is the impact of deregulation, which overlaps with our sample period for a few years in the early 2000s. Although we exclude ownership changes corresponding to divestitures, we conduct a robustness check by restricting our analysis to acquisitions after 2010. The results are reported in column 3 of table OA-13 and figure OA-19.

Matching difference-in-differences.—For the matching specification, we use the sample of first acquisitions that experience both subsidiary and parent change. We match each acquired generator with five never-acquired comparable units. For each unit, we first create a pool of potential control units that share the same fuel type and technology but operate in different markets (ISO) to prevent spillover effects. We then match these generators based on capacity and age using a least squares distance metric, with weights inversely proportional to each variable's standard deviation. Results are presented in column 5 of table OA-13 and figure OA-22.

Staggered difference-in-differences.—Recent developments in econometrics suggest that the two-way fixed effects difference-in-differences approach might produce a weighted average of all potential combinations of pairwise difference-in-differences estimators, where the control unit in the pair could be a unit that is treated at a different time (Callaway and Sant'Anna 2021; Goodman-Bacon 2021). To address this issue, we estimate cohort-specific treatment effects using the Callaway and Sant'Anna (2021) method. The results, reported in figure OA-21, are similar to our baseline results.

Placebo tests.—We use minority acquisitions as a placebo test against potential unobservable characteristics driving both acquisitions and efficiency changes. If such unobservables exist, they would likely influence minority acquisitions as well. The results, reported in column 6 of table OA-13, show no change in power plant efficiency following minority acquisitions.

VIII. Concluding Remarks

By reallocating resources between firms, acquisitions affect a significant portion of the economy. Despite their importance, there is limited systematic

evidence of their effects on productivity. This study provides detailed empirical analyses of the efficiency effects of ownership changes by examining a large sample of power plant acquisitions between 2000 and 2023 in the United States.

Our empirical results can be summarized into three principal findings. First, acquired plants experience, on average, a 2% increase in fuel efficiency within 5–18 months after acquisitions. This effect is more pronounced, rising to 5%, for acquisitions involving changes at both the subsidiary and parent-owner levels. Second, acquired generators tend to demonstrate improved operational performance: they produce more, increase their capacity utilization, and decrease their outage frequency and emission intensity. Finally, our evidence suggests that the new owners improve productivity by changing operational processes rather than by making capital investments.

Our findings draw on a large number of acquisitions in the power generation industry and high-frequency data on physical productivity. Using physical measurements in this homogeneous product setting allows us to disentangle the productivity effects from other potential merger effects, such as changes in market power, buyer power, or product quality. With high-frequency data, we can treat mergers as discrete events and compare generator productivity immediately before and after acquisitions. Finally, by aggregating evidence from numerous acquisitions, we have the statistical power to uncover the mechanisms that generate efficiency gains.

The results of this paper have important policy implications, as they provide direct input for evaluating the trade-off between market power and efficiency resulting from mergers. Our results present a mixed view of whether mergers generate efficiencies. On the one hand, we document that mergers in the electricity generation sector can generate efficiencies that are large and through a mechanism that could be considered specific to a merger. On the other hand, not all mergers necessarily generate efficiencies; we find no detectable average efficiency increase from parent-ownership changes, which tend to be larger and are less likely to influence plants' operations. In conclusion, the main message of our paper is that while efficiency effects in mergers should not be ruled out, they necessitate careful analysis tailored to the circumstances of each merger.

Data Availability

Code, data, and instructions needed to replicate the numerical results in the main text and online appendixes of this paper can be found in Demirer and Karaduman (2025) in the Harvard Dataverse, <https://doi.org/10.7910/DVN/GZMO6B>.

References

- Abito, Jose Miguel, Jin Soo Han, Jean-François Houde, and Arthur A. van Benthem. 2024. "Agency Frictions and Procurement: New Evidence from US Electricity Restructuring." *J. Indus. Econ.* 72 (1): 20–48.
- Abito, Jose Miguel, Christopher R. Knittel, Konstantinos Metaxoglou, and André Trindade. 2022. "The Role of Output Reallocation and Investment in Coordinating Environmental Markets." *Internat. J. Indus. Org.* 83:102843.
- Akerberg, Daniel A., Kevin Caves, and Garth Frazer. 2015. "Identification Properties of Recent Production Function Estimators." *Econometrica* 83 (6): 2411–51.
- Akey, Pat, and Ian Appel. 2021. "The Limits of Limited Liability: Evidence from Industrial Pollution." *J. Finance* 76 (1): 5–55.
- An, Yonghong, and Wei Zhao. 2019. "Dynamic Efficiencies of the 1997 Boeing–McDonnell Douglas Merger." *RAND J. Econ.* 50 (3): 666–94.
- Andonov, Aleksandar, and Joshua D. Rauh. 2023. "The Shifting Finance of Electricity Generation." Research Paper no. 4287123 (December), Graduate School Bus., Stanford Univ.
- Arora, Ashish, Sharon Belenzon, and Lia Sheer. 2021. "Matching Patents to Compustat Firms, 1980–2015: Dynamic Reassignment, Name Changes, and Ownership Structures." *Res. Policy* 50 (5): 104217.
- Ashenfelter, Orley, and Daniel Hosken. 2010. "The Effect of Mergers on Consumer Prices: Evidence from Five Mergers on the Enforcement Margin." *J. Law and Econ.* 53 (3): 417–66.
- Ashenfelter, Orley C., Daniel S. Hosken, and Matthew C. Weinberg. 2015. "Efficiencies Brewed: Pricing and Consolidation in the US Beer Industry." *RAND J. Econ.* 46 (2): 328–61.
- Asker, John, and Volker Nocke. 2021. "Collusion, Mergers, and Related Antitrust Issues." In *Handbook of Industrial Organization*, vol. 5, edited by Kate Ho, Ali Hortaçsu, and Alessandro Lizzeri, 177–279. Amsterdam: North-Holland.
- Atalay, Enghin. 2014. "Materials Prices and Productivity." *J. European Econ. Assoc.* 12 (3): 575–611.
- Atalay, Enghin, Ali Hortaçsu, and Chad Syverson. 2014. "Vertical Integration and Input Flows." *A.E.R.* 104 (4): 1120–48.
- Atalay, Enghin, Alan Sorensen, Christopher Sullivan, and Wanjia Zhu. 2024. "Product Repositioning by Merging Firms." *J. Indus. Econ.* 72 (2): 868–908.
- Berger, David W., Thomas Hasenzagl, Kyle F. Herkenhoff, Simon Mongey, and Eric A. Posner. 2023. "Merger Guidelines for the Labor Market." Working Paper no. 31147 (April), NBER, Cambridge, MA.
- Bhattacharya, Vivek, Gastón Illanes, and David Stillerman. 2024. "Merger Effects and Antitrust Enforcement: Evidence from US Consumer Packaged Goods." Working Paper no. 31123 (April), NBER, Cambridge, MA.
- Bitzan, John D., and Wesley W. Wilson. 2007. "Industry Costs and Consolidation: Efficiency Gains and Mergers in the US Railroad Industry." *Rev. Indus. Org.* 30 (2): 81–105.
- Blonigen, Bruce A., and Justin R. Pierce. 2016. "Evidence for the Effects of Mergers on Market Power and Efficiency." Working Paper no. 22750 (October), NBER, Cambridge, MA.
- Bloom, Nicholas, Raffaella Sadun, and John Van Reenen. 2012. "Americans Do IT Better: US Multinationals and the Productivity Miracle." *A.E.R.* 102 (1): 167–201.
- Bloom, Nicholas, and John Van Reenen. 2010. "Why Do Management Practices Differ across Firms and Countries?" *J. Econ. Perspectives* 24 (1): 203–24.

- Blundell, Richard, and James L. Powell. 2003. "Endogeneity in Nonparametric and Semi-parametric Regression Models." In *Advances in Economics and Econometrics*, edited by Mathias Dewatripont, Lars Peter Hansen, and Stephen J. Turnovsky, 312–57. Cambridge: Cambridge Univ. Press.
- Borenstein, Severin, and James B. Bushnell. 2022. "Do Two Electricity Pricing Wrongs Make a Right? Cost Recovery, Externalities, and Efficiency." *American Econ. J. Econ. Policy* 14 (4): 80–110.
- Borenstein, Severin, James Bushnell, and Erin Mansur. 2023. "The Economics of Electricity Reliability." *J. Econ. Perspectives* 37 (4): 181–206.
- Borrero, Miguel, Gautam Gowrisankaran, and Ashley Langer. 2024. "Ramping Costs and Coal Generator Exit." Working Paper no. 24-04 (August), Dept. Econ., Univ. Arizona.
- Bradley and Associates. 2017. *Powering America: The Economic and Workforce Contributions of the US Electric Power Industry*. Report, Nat. Rural Electric Cooperative Assoc. <https://www.electric.coop/wp-content/uploads/2017/08/Powering-America.pdf>.
- Braguinsky, Serguey, Atsushi Ohyama, Tetsuji Okazaki, and Chad Syverson. 2015. "Acquisitions, Productivity, and Profitability: Evidence from the Japanese Cotton Spinning Industry." *A.E.R.* 105 (7): 2086–119.
- Brand, Keith, Chris Garmon, and Ted Rosenbaum. 2023. "In the Shadow of Antitrust Enforcement: Price Effects of Hospital Mergers from 2009 to 2016." *J. Law and Econ.* 66 (4): 639–69.
- Brot-Goldberg, Zarek, Zack Cooper, Stuart V. Craig, and Lev Klarnet. 2024. "Is There Too Little Antitrust Enforcement in the US Hospital Sector?" *American Econ. Rev. Insights* 6 (4): 526–42.
- Bushnell, James B., Erin T. Mansur, and Frank A. Wolak. 2012. "Vertical Position and Price Effects of Mergers: The Exelon Constellation Merger." Working paper.
- Bushnell, James B., and Catherine Wolfram. 2005. "Ownership Change, Incentives and Plant Efficiency: The Divestiture of US Electric Generation Plants." CSEM Working Paper no. 140 (March), Cent. Study Energy Markets, Univ. California, Berkeley.
- . 2009. "The Guy at the Controls: Labor Quality and Power Plant Efficiency." In *International Differences in the Business Practices and Productivity of Firms*, edited by Richard B. Freeman and Kathryn L. Shaw, 79–102. Chicago: Univ. Chicago Press.
- Callaway, Brantly, and Pedro H. C. Sant'Anna. 2021. "Difference in Differences with Multiple Time Periods." *J. Econometrics* 225 (2): 200–30.
- Chen, Yanyou. 2024. "Network Structure and the Efficiency Gains from Mergers: Evidence from US Freight Railroads." Working paper, Univ. Toronto.
- Cicala, Steve. 2015. "When Does Regulation Distort Costs? Lessons from Fuel Procurement in US Electricity Generation." *A.E.R.* 105 (1): 411–44.
- . 2022. "Imperfect Markets versus Imperfect Regulation in US Electricity Generation." *A.E.R.* 112 (2): 409–41.
- Clark, Robert, and Mario Samano. 2022. "Incentivized Mergers and Cost Efficiency: Evidence from the Electricity Distribution Industry." *J. Indus. Econ.* 70 (4): 791–837.
- Craig, Stuart V., Matthew Grennan, and Ashley Swanson. 2021. "Mergers and Marginal Costs: New Evidence on Hospital Buyer Power." *RAND J. Econ.* 52 (1): 151–78.
- Dafny, Leemore, Kate Ho, and Robin S. Lee. 2019. "The Price Effects of Cross-Market Mergers: Theory and Evidence from the Hospital Industry." *RAND J. Econ.* 50 (2): 286–325.

- Davis, Lucas, and Catherine Hausman. 2016. "Market Impacts of a Nuclear Power Plant Closure." *American Econ. J. Appl. Econ.* 8 (2): 92–122.
- Davis, Lucas, and Catherine Wolfram. 2012. "Deregulation, Consolidation, and Efficiency: Evidence from US Nuclear Power." *American Econ. J. Appl. Econ.* 4 (4): 194–225.
- Davis, Rebecca J., J. Scott Holladay, and Charles Sims. 2022. "Coal-Fired Power Plant Retirements in the United States." *Environmental and Energy Policy and Econ.* 3 (1): 4–36.
- Davis, Steven J., John C. Haltiwanger, Kyle Handley, Ben Lipsius, Josh Lerner, and Javier Miranda. 2024. "The (Heterogeneous) Economic Effects of Private Equity Buyouts." *Management Sci.* 70 (3): 1491–513.
- De Loecker, Jan, and Chad Syverson. 2021. "An Industrial Organization Perspective on Productivity." In *Handbook of Industrial Organization*, vol. 4, edited by Kate Ho, Ali Hortaçsu, and Alessandro Lizzeri, 141–223. Amsterdam: North-Holland.
- De Loecker, Jan, and Frederic Warzynski. 2012. "Markups and Firm-Level Export Status." *A.E.R.* 102 (6): 2437–71.
- Demirer, Mert, and Omer Karaduman. 2025. "Replication Data for: 'Do Mergers and Acquisitions Improve Efficiency? Evidence from Power Plants.'" Harvard Dataverse, <https://doi.org/10.7910/DVN/GZMO6B>.
- DOJ (US Department of Justice) and FTC (Federal Trade Commission). 2023. "Merger Guidelines." Policy document. https://www.ftc.gov/system/files/ftc_gov/pdf/2023_merger_guidelines_final_12.18.2023.pdf.
- Dranove, David, and Richard Lindrooth. 2003. "Hospital Consolidation and Costs: Another Look at the Evidence." *J. Health Econ.* 22 (6): 983–97.
- EIA (US Energy Information Administration). 2015. "Analysis of Heat Rate Improvement Potential at Coal-Fired Power Plants." Report. <https://www.eia.gov/analysis/studies/powerplants/heatrate/pdf/heatrate.pdf>.
- . 2020. "Annual Energy Outlook 2020." Report. <https://www.eia.gov/outlooks/aeo/pdf/AEO2020%20Full%20Report.pdf>.
- Eliason, Paul J., Benjamin Heebsh, Ryan C. McDevitt, and James W. Roberts. 2020. "How Acquisitions Affect Firm Behavior and Performance: Evidence from the Dialysis Industry." *Q.J.E.* 135 (1): 221–67.
- Elliott, Jonathan T., Georges Vivien Hounghonon, Marc Ivaldi, and Paul Scott. 2025. "Market Structure, Investment, and Technical Efficiencies in Mobile Telecommunications." *J.P.E.* 133 (5): 1401–59.
- Emerson Process Management. 2016. "Using Automation to Improve Plant Heat Rate." Report. <https://www.emerson.com/documents/automation/white-paper-using-automation-to-improve-plant-heat-rate-rosemount-en-178352.pdf>.
- Environmental Defense Fund. 2017. "Improving Heat Rate on Combined Cycle Power Plants." Report. https://www.andoverttechnology.com/wp-content/uploads/2021/03/C_18_EDF_FINAL.pdf.
- EPA (US Environmental Protection Agency). 2018. "Overview of the Clean Power Plan." Fact sheet. <https://archive.epa.gov/epa/cleanpowerplan/fact-sheet-overview-clean-power-plan.html>.
- EPRI (Electric Power Research Institute). 2014. "Range and Applicability of Heat Rate Improvements." Technical update. <https://www.epri.com/research/products/000000003002003457>.
- . 2022. "2022 EPRI Heat Rate Improvement Conference Proceedings." Proceedings of the EPRI Heat Rate Improvement Conference, Charleston, SC, February 7–9. <https://www.epri.com/research/programs/113126/results/3002022356>.

- Fabrizio, Kira R., Nancy L. Rose, and Catherine D. Wolfram. 2007. "Do Markets Reduce Costs? Assessing the Impact of Regulatory Restructuring on US Electric Generation Efficiency." *A.E.R.* 97 (4): 1250–77.
- Fan, Ying, and Chenyu Yang. 2025. "Estimating Discrete Games with Many Firms and Many Decisions: An Application to Merger and Product Variety." *J.P.E.* 133 (5): 1401–59.
- Farrell, Joseph, and Carl Shapiro. 1990. "Horizontal Mergers: An Equilibrium Analysis." *A.E.R.* 80 (1): 107–26.
- . 2010. "Antitrust Evaluation of Horizontal Mergers: An Economic Alternative to Market Definition." *B.E. J. Theoretical Econ.* 10 (1): 9.
- FERC (Federal Energy Regulatory Commission). 2012. "Analysis of Horizontal Market Power under the Federal Power Act: Order Reaffirming Commission Policy and Terminating Proceeding. 138 FERC 61,109." Docket no. RM11–14–000, FERC, Washington, DC.
- Fitzgerald, Patrick, and Jenifer Gelorme. 2015. "Thermal Cycling, Air In-Leakage, Impact on Heat Rate, Emissions and Cost of Compliance." Paper presented at the EPRI Heat Rate Improvement Conference, Houston, TX, February 3–5.
- Focarelli, Dario, and Fabio Panetta. 2003. "Are Mergers Beneficial to Consumers? Evidence from the Market for Bank Deposits." *A.E.R.* 93 (4): 1152–72.
- Foster, Lucia, John Haltiwanger, and Chad Syverson. 2008. "Reallocation, Firm Turnover, and Efficiency: Selection on Productivity or Profitability?" *A.E.R.* 98 (1): 394–425.
- Geurts, Karen, and Johannes Van Biesebroeck. 2019. "Employment Growth Following Takeovers." *RAND J. Econ.* 50 (4): 916–50.
- Gibbons, Robert, and Rebecca Henderson. 2012. "What Do Managers Do? Exploring Persistent Performance Differences among Seemingly Similar Enterprises." In *The Handbook of Organizational Economics*, edited by Robert Gibbons and John Roberts, 680–731. Princeton, NJ: Princeton Univ. Press.
- GMI (Global Market Intelligence). 2024. *US and Canada Energy Coverage Letter*. https://www.spglobal.com/marketintelligence/en/documents/mi_nonfin_387622_energy-cv-us-and-canada-energy-coverage_letter-final.pdf.
- Goodman-Bacon, Andrew. 2021. "Difference-in-Differences with Variation in Treatment Timing." *J. Econometrics* 225 (2): 254–77.
- Gowrisankaran, Gautam, Ashley Langer, and Wendan Zhang. 2025. "Policy Uncertainty in the Market for Coal Electricity: The Case of Air Toxics Standards." *J.P.E.* 133 (5): 1460–502.
- Grieco, Paul, Joris Pinkse, and Margaret Slade. 2018. "Brewed in North America: Mergers, Marginal Costs, and Efficiency." *Internat. J. Indust. Org.* 59:24–65.
- Haldi, John, and David Whitcomb. 1967. "Economies of Scale in Industrial Plants." *J.P.E.* 75 (4, pt. 1): 373–85.
- Harrison, Teresa D. 2011. "Do Mergers Really Reduce Costs? Evidence from Hospitals." *Econ. Inquiry* 49 (4): 1054–69.
- Hausman, Catherine. 2014. "Corporate Incentives and Nuclear Safety." *American Econ. J. Econ. Policy* 6 (3): 178–206.
- Hempling, Scott. 2018. "Inconsistent with the Public Interest: FERC's Three Decades of Deference to Electricity Consolidation." *Energy Law J.* 39:233.
- Hortaçsu, Ali, Fernando Luco, Steven L. Puller, and Dongni Zhu. 2019. "Does Strategic Ability Affect Efficiency? Evidence from Electricity Markets." *A.E.R.* 109 (12): 4302–42.
- Hortaçsu, Ali, and Steven L. Puller. 2008. "Understanding Strategic Bidding in Multi-unit Auctions: A Case Study of the Texas Electricity Spot Market." *RAND J. Econ.* 39 (1): 86–114.

- Hortaçsu, Ali, and Chad Syverson. 2007. "Cementing Relationships: Vertical Integration, Foreclosure, Productivity, and Prices." *J.P.E.* 115 (2): 250–301.
- Jeziorski, Przemysław. 2014. "Estimation of Cost Efficiencies from Mergers: Application to US Radio." *RAND J. Econ.* 45 (4): 816–46.
- Jha, Akshaya. 2020. "Dynamic Regulatory Distortions: Coal Procurement at US Power Plants." Working paper, Carnegie Mellon Univ.
- Jha, Akshaya, and Gordon Leslie. 2023. "Start-Up Costs and Market Power: Lessons from the Renewable Energy Transition." Working paper.
- Joskow, Paul L., Douglas R. Bohi, and Frank M. Gollop. 1989. "Regulatory Failure, Regulatory Reform, and Structural Change in the Electrical Power Industry." *Brookings Papers Econ. Activity* 20:125–208.
- Jovanovic, Boyan, and Peter L. Rousseau. 2002. "The Q-Theory of Mergers." *A.E.R.* 92 (2): 198–204.
- Kalnins, Arturs, Luke Froeb, and Steven Tschantz. 2017. "Can Mergers Increase Output? Evidence from the Lodging Industry." *RAND J. Econ.* 48 (1): 178–202.
- Knittel, Christopher R. 2002. "Alternative Regulatory Methods and Firm Efficiency: Stochastic Frontier Evidence from the US Electricity Industry." *Rev. Econ. and Statis.* 84 (3): 530–40.
- Kulick, Robert B. 2017. "Ready-to-Mix: Horizontal Mergers, Prices, and Productivity." CES Paper no. CES-WP-17-38 (April), Cent. Econ. Stud., US Census Bur., Washington, DC.
- Kwoka, John. 2014. *Mergers, Merger Control, and Remedies: A Retrospective Analysis of US Policy*. Cambridge, MA: MIT Press.
- Kwoka, John, and Michael Pollitt. 2010. "Do Mergers Improve Efficiency? Evidence from Restructuring the US Electric Power Sector." *Internat. J. Indus. Org.* 28 (6): 645–56.
- La Forgia, Ambar. 2024. "The Impact of Management on Clinical Performance: Evidence from Physician Practice Management Companies." *Management Sci.* 69 (8): 4646–67.
- Lewis, Matthew S., and Kevin E. Pflum. 2017. "Hospital Systems and Bargaining Power: Evidence from Out-of-Market Acquisitions." *RAND J. Econ.* 48 (3): 579–610.
- Luco, Fernando, and Guillermo Marshall. 2020. "The Competitive Impact of Vertical Integration by Multiproduct Firms." *A.E.R.* 110 (7): 2041–64.
- Macchiavello, Rocco, and Ameet Morjaria. 2022. "Acquisitions, Management, and Efficiency in Rwanda's Coffee Industry." Working Paper no. 30230 (July), NBER, Cambridge, MA.
- MacKay, Alexander, and Ignacia Mercadal. 2024. "Do Markets Reduce Prices? Evidence from the US Electricity Sector." Working paper.
- Malmendier, Ulrike, Enrico Moretti, and Florian S. Peters. 2018. "Winning by Losing: Evidence on the Long-Run Effects of Mergers." *Rev. Financial Studies* 31 (8): 3212–64.
- Marschak, Jacob, and William H. Andrews. 1944. "Random Simultaneous Equations and the Theory of Production." *Econometrica* 12 (3/4): 143–205.
- Midrigan, Virgiliu, and Daniel Yi Xu. 2014. "Finance and Misallocation: Evidence from Plant-Level Data." *A.E.R.* 104 (2): 422–58.
- Miller, Nathan H., and Matthew C. Weinberg. 2017. "Understanding the Price Effects of the MillerCoors Joint Venture." *Econometrica* 85 (6): 1763–91.
- Morris, John R., and Daniel Oska. 2008. "The Likely Effect of the Proposed Exelon-PSEG Merger on Wholesale Electricity Prices." *Electricity J.* 21 (1): 45–54.
- Moshary, Sarah, and Cailin Slattery. 2024. "Consolidation and Political Influence in the Auto Retail Industry." Working paper, Haas School Bus., Univ. California, Berkeley.

- Nguyen, Sang V., and Michael Ollinger. 2006. "Mergers and Acquisitions and Productivity in the US Meat Products Industries: Evidence from the Micro Data." *American J. Agricultural Econ.* 88 (3): 606–16.
- Niefer, Mark J. 2012. "Explaining the Divide Between DOJ and FERC on Electric Power Merger Policy." *Energy Law J.* 33 (2): 505–36.
- Nocke, Volker, and Michael D. Whinston. 2022. "Concentration Thresholds for Horizontal Mergers." *A.E.R.* 112 (6): 1915–48.
- PacifiCorp Energy. 2016. "Fossil Fuel Heat Rate Improvement Plan." <https://pscdocs.utah.gov/electric/16docs/1603519/275489PacifiCorpFossilFuelHeatRateImprovPlan4-29-2016.pdf>.
- Prager, Elena, and Matt Schmitt. 2021. "Employer Consolidation and Wages: Evidence from Hospitals." *A.E.R.* 111 (2): 397–427.
- Rhodes-Kropf, Matthew, and David T. Robinson. 2008. "The Market for Mergers and the Boundaries of the Firm." *J. Finance* 63 (3): 1169–211.
- Röller, Lars-Hendrik, Johan Stennek, and Frank Verboven. 2006. "Efficiency Gains from Mergers." In *European Merger Control: Do We Need an Efficiency Defence?*, edited by Fabienne Ilzkovitz and Roderick Meiklejohn, chap. 3. Brussels: European Commission.
- Rosaia, Nicola. 2020. "Competing Platforms and Transport Equilibrium: Evidence from New York City." Working paper, Dept. Econ., Harvard Univ.
- Sargent and Lundy. 2009. *Coal-Fired Power Plant Heat Rate Reductions*. Report, Perrin Quarles on behalf of the US Environmental Protection Agency, Washington, DC.
- Schmitt, Matt. 2017. "Do Hospital Mergers Reduce Costs?" *J. Health Econ.* 52:74–94.
- Staudt, James E., and Jennifer Macedonia. 2014. "Evaluation of Heat Rates of Coal Fired Electric Power Boilers." Paper presented at the Power Plant Pollutant Control MEGA Symposium, August 19–21, Baltimore, MD.
- Syverson, Chad. 2011. "What Determines Productivity?" *J. Econ. Literature* 49 (2): 326–65.
- Weber, Roberto A., and Colin F. Camerer. 2003. "Cultural Conflict and Merger Failure: An Experimental Approach." *Management Sci.* 49 (4): 400–415.
- Weinberg, Matthew. 2008. "The Price Effects of Horizontal Mergers." *J. Competition Law and Econ.* 4 (2): 433–47.
- Whinston, Michael D. 2007. "Antitrust Policy toward Horizontal Mergers." *Handbook of Industrial Organization*, vol. 3, edited by M. Armstrong and R. Porter, 2369–440. Amsterdam: North-Holland.
- Williamson, O. 1968. "Economies as an Antitrust Defense: The Welfare Trade-offs." *A.E.R.* 58 (1): 18–36.
- Wolak, Frank, and Shawn McRae. 2007. "Merger Analysis in Restructured Electricity Supply Industries: The Proposed PSEG and Exelon Merger (2006)." Working paper, Dept. Econ., Stanford Univ.